BIOLOGY

CLASS - 10

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QR CODE TEAM

D1H1W7

Published by the Government of Telangana, Hyderabad.

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Grow by Education
Behave Humbly

First Published 2014

New Impression 2015, 2016, 2017, 2018, 2019, 2020, 2021

This Book has been printed on 70 G.S.M. S.S. Map litho,
Title Page 200 G.S.M. White Art Card

Free Distribution by Government of Telangana 2021-22

Printed in India
at the Telangana Govt. Text Book Press,
Mint Compound, Hyderabad,
Telangana.

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Intro ...

We believe that class-10 education is a key aspect of school education and a turning point in student's life. The present tenth class Science textbook in your hands is developed in accordance with the National and State Curriculum Framework and the Right to Education Act. This book helps the student to review various concepts that were learned through the learning experiences provided in the school and to get comprehensive knowledge about these conecepts. The lessons in the textbooks are presented in such way that they help in preparing the student for competitive examinations and also to prepare him/her for intermediate education.

The new science textbooks are specially designed with suitable pedagogy in tune with Continuous Comprehensive Evaluation (CCE) which we are right now implementing in school education. These textbooks help the teacher to assess students learning during teaching learning processes. They facilitate effective learning of various concepts of science in scientific method apart from getting knowledge about concepts. It is essential to complete the syllabus in the stipulated time as students have to write the Board Examination in class-10. But don't forget that completion of syllabus means making the students understand the concepts and trying to achieve the learning competencies. It is mandatory on the part of teacher to implement teaching strategies like, making the student read the content of the textbook, discuss, analyse, do lab activity, go for field trips, prepare reports, etc. Teacher must take special care to avoid the practise of memorisation of scientific information from guides and question banks.

The classroom teaching of science must be in such a way that it encourages children to think and work scientifically. It must also enhance love towards nature. It should enable one to comprehend and appreciate the laws governing nature and immense diversity in it. Scientific learning is not just disclosing new things. It is also necessary to step forward without interrupting the interrelationship and interdependency along with understanding of the nature's intrinsic principles.High School children possess cognitive capacity of comprehending the nature and characteristics of the transforming world surrounding them and analyze abstract concepts.

At this level, we cannot quench their sharp thinking capability with the dry teaching of mere equations and theoretic principles. For that, we should create a learning environment in the classroom which provides an opportunity for them to apply the scientific knowledge, explore multiple alternatives in solving problems and establish new relations.

Scientific learning is not just confined to the four walls of classroom. It has a definite connection to lab and field as well. Therefore, there is a lot of importance to field experience/ experiments in science teaching.

There is a great need for compulsory implementation of instructions of the National Curriculum Framework- 2005 which emphasizes linking of the science teaching with local environment. The Right to Education Act- 2009 also suggested that priority should be given to the achievement of learning competencies among children. Likewise, science teaching should be in such a way that it would help cultivate a new generation with scientific thinking.

The key aspect of science teaching is to make the children understand the thinking process of scientists and their efforts behind each discovery. The State Curriculum Framework- 2011 stated thatchildren should be able to express their own ideas and opinions on various aspects. These Science Text Books are prepared to meet the set standards of the SCF and thus assist children in becoming self-reliant researchers capable of thinking intensely in scientific terms.

The new textbooks are developed to achieve desired academic standards. So teachers should develop various teaching learning strategies to make their students to achieve class specific academic standards. We should avoid rote learning methods for the successful implementation of Continuous Comprehensive Evaluation (CCE). It is very important to know more about different methods to assess students' progress by summative and formative evaluation. The new textbooks reflect Continuous Comprehensive Evaluation and teaching method with respect to discussed concepts. This is very useful to teachers and students.

In the new textbooks, the design of concepts and activities help in achieving the specified academic standards. Teachers need to plan appropriate teaching strategies to improve the academic standards among the students by the end of teaching the lesson. For effective implementation of continuous comprehensive evaluation the teaching must move away from the methods of memorisation of concepts. There is a need for teachers to have a good understanding of the methods of evaluation which help them in assessing the progress of children in a constructive and comprehensive way. The new textbooks are not confined to simply provide necessary informations about concepts. Instead they focus on the new teaching strategies and evaluation techniques which are very important for both teachers and students.

We thank the Vidya Bhawan Society, Rajasthan for their cooperation in designing these new text books,the writers for preparing the lessons, Smt. Pulipati Parameswari who helped in translation, the editors for checking the textual matters and the DTP group for composing the textbook. We invite suggestions from educationists, teachers, parents, students and others to make this book more meaningful.

Teachers play a pivotal role in children's comprehensive use of the text book. We hope, teachers will make their best efforts for proper utilization of the text book so as to inculcate scientific thinking among children and inspire them to be great scientists.

Director,
SCERT, Hyderabad

Dear teachers...

New Science Text Books are prepared in such a way that they develop children's observation power and research enthusiasm. The official documents of National and State Curriculum Frame works and the Right to Education Act are aspiring to bring grass root changes in science teaching. These textbooks are adopted in accordance with such an aspiration. Hence, science teachers need to adapt to the new approach in their teaching.

In view of this, let us observe certain **Do**s and **Don't**s:

- There is an immediate need to discard the practices adopted in the schools on a false belief that teaching of 10^{th} class means preparing the children for public examination. In 10^{th} class the teaching learning process should focus on achieving the academic standards rather than winning race of scoring marks.
- Avoid practices like using guides and question banks, asking the children to read only important questions, focussing on lessons which help in scoring more marks.
- Read the lesson thoroughly before you start teaching and ask the children to read the text. Then initiate a discussion to make the children understand the concepts in the lessons.
- Encourage children to express their own views and ideas while writing the answers. Give weightage to such type of writing in examination.
- Some instructions are given in the textbook regarding the collection of certain information by the teacher. Collect such information and make it available to students.
- In public examination, the weightage will be given to all aspects of the syllabus. Except foreword of the textbook everything else must be treated as a part of the curriculum.
- Textual concepts are presented in two ways: one as the classroom teaching and the other as the laboratory performance.
- Lab activities are part and parcel of a lesson. So, teachers must make the children conduct all such activities during the lesson itself, but not separately.
- Teachers are advised to follow the following teaching steps while transacting lessons-mind mapping, reading lesson and identifyng new words by children, performing activities, demonstration and discussion, conclusion and evaluation.
- In the text, some special activities are presented as box items: 'think and discuss, let us do, conduct interview, prepare report, display in wall magazine, participate in Theatre Day, do field observation, organize special days'. To perform all of them is compulsory.
- The abbreviation (A.S.) given at the end of each question in the section "Improve your learning" indicates academic standard.
- Collect information of relevant website addresses and pass on to students so that they can utilize internet services for learning science on their own.
- Plan and execute activities like science club, elocution, drawing, writing poetry on science, making models *etc.*to develop positive attitude among children about environment, biodiversity, ecological balance *etc.*

- As a part of continuous comprehensive evaluation, observe and record children's learning abilities during various activities conducted in classroom, laboratory and field.
- Teaching learning strategies and the expected learning outcomes, have been developed class wise and subject-wise based on the syllabus and compiled in the form of a Hand book to guide the teachers and were supplied to all the schools. With the help of this Hand book the teachers are expected to conduct effective teaching learning processes and ensure that all the students attain the expected learning outcomes.

We believe, you must have realized that the learning of science and scientific thinking are not mere drilling of the lessons but, in fact, a valuable exercise in motivating the children to explore for solutions to the problems systematically and preparing them to meet life's challenges properly.

Dear Students...

Learning science does not mean scoring good marks in the subject. Competencies like thinking logically and working systematically, learned through it,have to be practiced in daily life. To achieve this, instead of memorizing the scientific theories by rote, one must be able to study them analytically. That means, in order to understand the concepts of science, you need to proceed by discussing, describing, conducting experiments to verify, making observations, confirming with your own ideas and drawing conclusions. This textbook helps you to learn in that way.

What you need to do to achieve such things:

- In 10^{th} class the range of concepts is wide. So go through each lesson thoroughly before the teacher actually deals with it.
- Note down the points you came across so that you can grasp the lesson better.
- Think of the principles in the lesson. Identify the concepts you need to know further, to understand the lesson in depth.
- Do not hesitate to discuss analytically about the questions given under the sub-heading 'Think and Discuss' with your friends or teachers.
- You may get some doubts while conducting an experiment or discussing about a lesson. Express them freely and clearly.
- Plan to implement experiment/lab periods together with teachers, to understand the concepts clearly. While learning through the experiments you may come to know many more things.
- Observe how each lesson is helpful to conserve nature. Put what you learnt into practice.
- Analyse how each teaching point has relation with daily life and discuss the things you learned in your science class with farmers, artisans *etc.*
- Work as a group during interviews and field trips. Preparing reports and displaying them is a must. Discuss on the report prepared.
- List out the observations regarding each lesson to be carried through internet, school library and laboratory.
- Whether in note book or exams, write analytically, expressing your own opinions.
- Read books related to your text book, as many as you can.
- You organize yourself the Science Club programs in your school.
- Observe problems faced by the people in your locality and find out what solutions you can suggest through your science classroom.

ACADEMIC STANDARDS

S.No.	Academic Standard	Explanation
1.	Conceptual understanding	Children are able to explain, cite examples, give reasons, compare and write differences, explain the process of given concepts in the textbook. Children are able to develop their own brain mappings.
2.	Asking questions and making hypothesis	Children are able to ask questions to understand, to clarify the concepts and to participate in discussions. They are able to make hypothesis on experimental results and given issues.
3.	Experimentation and field investigation.	To understand given concepts in the textbook, children are able to do experiments on their own. They are able to arrange the experimental materials, note their observations, collect alternate expermental materials, take precautions, participate in field investigation and make reports on them.
4.	Information skills and Projects	Children are able to collect information (by using interviews, checklist, questionaire) and analyses systematically. They are able to conduct their own project works.
5.	Communication through drawing, model making	Children are able to explain their conceptual understanding by drawing figures labelling, describing the parts and making models. They are able to plot graphs by using given information or collected data.
6.	Appreciation and aesthetic sense, values	Children are able to appreciate man power and nature, and have aesthetic sense towards nature. They are also able to follow constitutional values.
7.	Application to daily life, concern to bio diversity.	Children are able to utilize scientific concept to face their daily life situations. They are able to show concern towards bio diversity.

INDEX

OUR NATIONAL ANTHEM

- Rabindranath Tagore

Jana-gana-mana-adhinayaka, jaya he
Bharata-bhagya-vidhata.
Punjab-Sindh-Gujarat-Maratha
Dravida-Utkala-Banga
Vindhya-Himachala-Yamuna-Ganga
Uchchhala-jaladhi-taranga.
Tava shubha name jage,
Tava shubha asisa mage,
Gahe tava jaya gatha,
Jana-gana-mangala-dayaka jaya he
Bharata-bhagya-vidhata.
Jaya he! jaya he! jaya he!
Jaya jaya jaya, jaya he!!

PLEDGE

- Pydimarri Venkata Subba Rao

"India is my country; all Indians are my brothers and sisters.
I love my country, and I am proud of its rich and varied heritage.

I shall always strive to be worthy of it.

I shall give my parents, teachers and all elders respect,
and treat everyone with courtesy. I shall be kind to animals.

To my country and my people, I pledge my devotion.

In their well-being and prosperity alone, lies my happiness."

Chapter 1

Nutrition

Food is needed by all living organisms mainly for growth and repair. Several organisms need food to maintain body temperature as well. A variety of substances are taken as food by organisms from single celluar organisms like amoeba to the complex multicellular organisms like the human body. Even within the human body the cells require a wide variety of substances as food to carryout their functions. The mode of acquiring food varies from organism to organism.

You have studied in the previous classes about how different organisms get their food. Let us recall them.

- *What are autotrophs? How do autotrophs get their food?*
- *What are heterotrophs? How do heterotrophs get their food?*

Let us study about autotrophic and heterotrophic modes of nutrition and find out why most plants are called as autotrophs.

Autotrophic Nutrition

We know that autotrophs are the organisms capable of using light energy to synthesize chemical compounds. They acquire nutrients like mineral salts and water from the soil as well as some gases from the air. They are capable of producing complex compounds like carbohydrates, proteins and lipids from these very simple substances. These compounds produced by autotrophic plants are utilized for providing energy to most of the living organisms.

Most of the food that we eat is obtained from plants. Even if we depend on animal products, we would find that those animals usually depend on plants for their food. But how do plants carry out their life processes?

Scientists have been working for centuries to find out how plants carry out their life processes. We know that among all the life processes, the process of photosynthesis makes plants "The universal food providers" for all living organisms.

You have studied something about photosynthesis in your earlier classes. Investigations into plant life processes might have started long ago. Actual experimentations to know about plant needs have been recorded from 17th century. A scientist named Van Helmont established the role of water by an experiment conducted by him for a period of 5 years. It took humankind another 300 years to arrive at the present definition of photosynthesis.

- *Can you name some raw materials needed for photosynthesis?*
- *What could be the end products of the process of photosynthesis?*

Let us study the process of photosynthesis in detail.

Photosynthesis

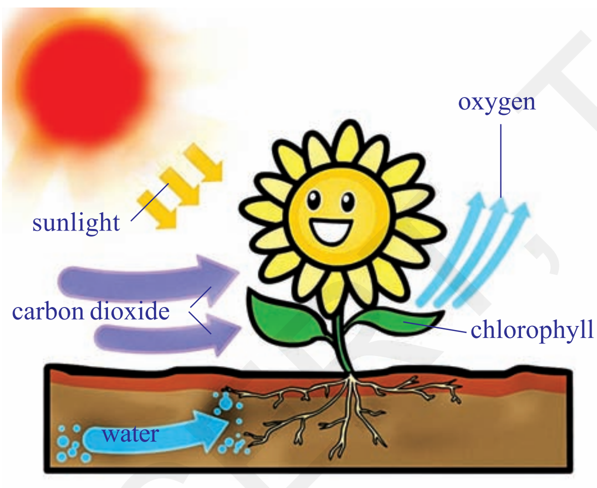


Fig-1: Photosynthesis

Photosynthesis is the process by which plants containing the green pigment 'chlorophyll' build up complex organic molecules from simple inorganic ones, using sun light as an energy source. The process of photosynthesis is very complex. Several reactions take place in this process and intermediate compounds are also formed. Scientists had tried to formulate a simple equation to indicate the process. The equation used widely is related to the one proposed by C.B. Van Niel in 1931. His opinion was, "For each molecule of carbohydrate formation, one molecule of carbon dioxide and two molecules of water are required. In this process, along with carbohydrate one molecule each of oxygen and water are produced".

$$CO_2 + 2H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} CH_2O + H_2O + O_2$$

What would be the reaction to show that glucose ($C_6H_{12}O_6$) is being synthesized? Write down a balanced equation to show this.

(Refer the lessons Chemical equations, Carbon and its compounds in your textbook of physical science class X).

C.B. Van Niel first worked on purple sulphur bacteria and found light plays specific role in photosynthesis. Instead of H_2O he used H_2S as a starting material. Here no oxygen is liberated during photosynthesis, instead, it is sulphur. Later he envisioned a similar process of photosynthesis, in plants. When glucose was observed to be a product of photosynthesis, the following equation was proposed:

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6H_2O + 6O_2$$

It is known that plants synthesize carbohydrates, the smaller and simpler ones first and from these, the more complex ones like starch and cellulose. Plants are also capable of synthesizing all other compounds like proteins, fats etc.

Animals are not capable of synthesizing carbohydrates and they have to depend on plants for the same.

Can we state that photosynthesis is the basic energy source for most of the living world? Why, why not?

Let us study how plants prepare carbohydrates through photosynthesis.

Activity-1

Presence of starch (a type of carbohydrate) in leaves

Let us take a leaf from a plant (soft and thin leaves and well exposed to sunlight).

Arrange apparatus as shown in Fig-2(a). Take methylated spirit in a test tube and put the leaf in it.

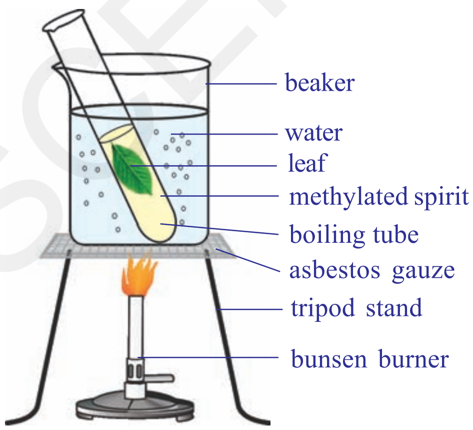


Fig-2(a): boiling the leaf in methylated spirit

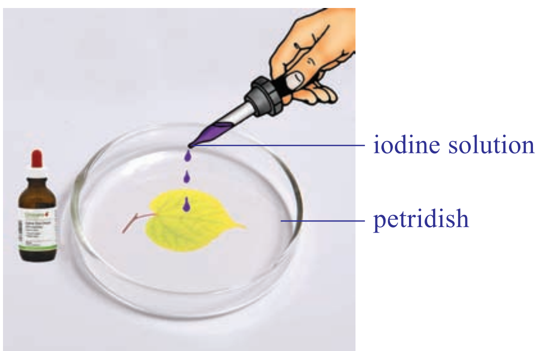


Fig-2(b): Iodine test

First boil the leaf in water then transfer this leaf into a test tube containing methylated spirit. Boil this tube in a water bath. Chlorophyll dissolves in spirit and the leaf becomes pale, due to removal of chlorophyll.

Take the leaf carefully from test tube by using a brush.

Spread the leaf in a petridish and add a few drops of iodine/ betadine solution on it. Observe the leaf.

- *What do you observe?*

The presence of starch will be indicated by a bluish black colour.

- *Do you think sunlight has a role in formation of starch?*

Factors (Materials) essential for the process of Photosynthesis

- *What are the materials that you think would be essential for the synthesis of carbohydrates in the process of photosynthesis? (Hint: see the equation proposed by Van Niel)*
- *Do you think the equation tells us about all the materials involved?*

Let us study how scientists experimented to find out about some of the materials required for the process of photosynthesis.

Water and Photosynthesis

In class VII we have already studied how Von Helmont found that the water is essential for the increase of plant mass.

He did not know about photosynthesis then. Later, it was found that increase in plant body mass or material occurred due to the process of photosynthesis. We shall study more about it in the following sections.

Air and Photosynthesis

Let us discuss an experiment on photosynthesis. This one helps us to find out about the role of air in the process of photosynthesis. It is interesting to learn about the experiment which is one of the several milestones in the gradual development of our understanding of Photosynthesis.

Joseph Priestley (1733-1804) in 1770 performed a series of experiments that revealed the essential role of air on the growth of green plants (photosynthesis was still not known to scientists at that time). You may recall, Oxygen was discovered by Priestley in 1774. The name oxygen was coined later by Lavoisier in the year 1775. Priestley observed that a candle burning in a closed bell jar, soon gets extinguished. Similarly, a mouse would soon suffocate in a closed space of the bell jar. He concluded that a burning candle or an animal, both somehow, damage air. When he placed a mint plant in the same bell jars, he found that the mouse stayed alive and the candle when lighted from outside continued burning. Priestley hypothesized as follows: 'Plants restore the air what breathing animals and burning candles remove'.

Do you find any relationship between candle, rat, mint plant ? Discuss.

Priestley's experiment confirmed that gaseous exchange was going on and plants were giving out a gas that supported burning and was essential for the survival of animals.

But how do plants take in air and utilize carbon dioxide for photosynthesis and oxygen for respiration?

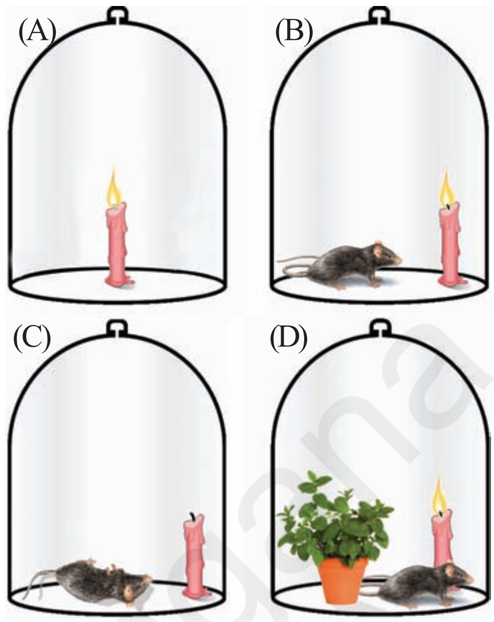


Fig-3: Priestley experiment

Massive amounts of gaseous exchange occur through the stomata (usually present in leaves) as long as they are open .While plants also carry on gaseous exchange through loose tissues in the stems, roots etc.

Activity-2

Carbondioxide is necessary for Photosynthesis (Mohl's half leaf experiment)

We need a destarched plant to start with. We need to keep the plant in the dark for nearly three days to remove the starch (destarch). By placing the plant in dark the starch is removed from the leaves.

Arrange the apparatus as shown in the Fig-4.

- Take a wide mouthed transparent bottle.

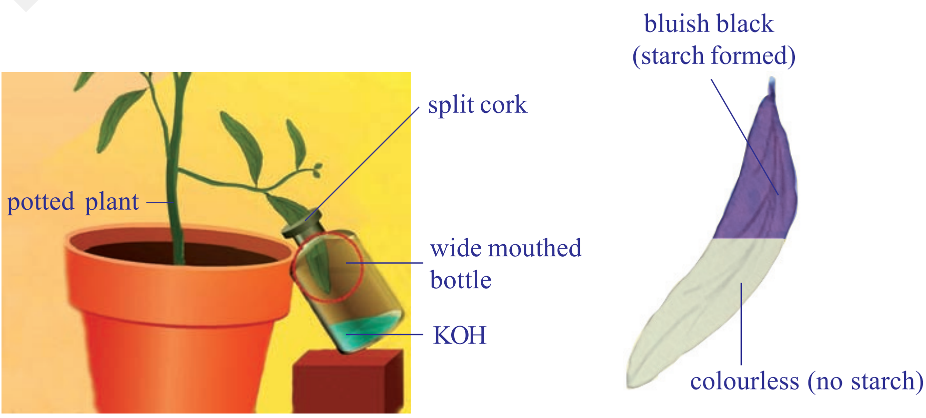


Fig-4: Mohl's half leaf experiment

- Put potassium hydroxide (KOH) pellets or potassium hydroxide solution in the bottle. Potassium hydroxide absorbs carbon dioxide.
- Insert splitted cork in the mouth of the bottle.
- Insert one of the leaf of destarched plant (through a split cork) into transparent bottle containing potassium hydroxide pellets/potassium hydroxide solution.
- Leave the plant in sunlight.
- After a few hours, test this leaf and any other leaf of this plant for presence of starch, as mentioned in activity-1.
- The leaf part which was exposed to the atmospheric air and light becomes bluish-black, and the one inside the bottle containing potassium hydroxide which absorbs carbon dioxide in the bottle remains colourless. This proves that carbon dioxide is necessary for photosynthesis.

Why was the plant kept in dark and then in sun light ?

Why did we test two leaves in this experiment?

We have so far discussed the role of water and gases in the process of photosynthesis. Scientist who had been working on these lines had observed some other factors that affect the process of photosynthesis.

Light and Photosynthesis

In Priestley's time, scientists didn't understand about energy, but later on much was discovered about it. If energy is released when carbon dioxide and water is formed by combining oxygen with carbon and hydrogen, then what about the reverse?. What about forming oxygen again and putting it back in the air. Eventually, scientists learned that the energy situation would also reverse. Oxygen formation would use up energy. That means if plants form oxygen they have to get energy to make it possible. Where did the energy come from?

A Dutch scientist, Jan Ingenhousz (1730-1799), found the answer. He studied the way in which plants form oxygen. In 1779, he noticed that it happened only in the presence of light. In an experiment with the aquatic plant, Hydrilla, he observed that in bright sunlight, small bubbles formed around the green parts while in the dark they did not form. He also found that the gas present in the bubbles was oxygen.

It was further confirmed when Engelman in the early 20th century ingeniously detected the point of maximum rate of photosynthesis. He used a strand of algae and exposed it to different colours of light (the colours that we see in a rainbow) Then he used oxygen sensitive bacteria and found that they crowd around areas illuminated with red and blue rays of light. This led to more studies on effect of light on photosynthesis, the role of different coloured compounds called pigments in plants and the utilization of light energy.

Lab Activity

Experiment to observe oxygen is evolved during photosynthesis in the presence of light

Materials required : Beaker (2), funnel (2), test tubes (2), *Hydrilla* (any other acquatic submerged plant), black paper, bucket-full of water, incense stick, match box.

Procedure

1. Arrange the apparatus as shown in the Fig-5 and make two identical sets with water plant (like *Hydrilla*).
2. Take the whole set under a bucket of water and invert a water filled test tube over the stem of funnel (This will help to retain the water column in the test tube).
3. Take the set up out of the bucket and keep it under sunlight.
4. Arrange the second set in the same way, cover it with black paper/ cloth. Keep them under shade.

You would see that in place of water there is air that fills in the set up kept in sun. It is actually a gas that will be collected in the test-tube. Observe the other set up kept in dark. Is there any difference in the amount of gas collected?

Test the gas in the test-tube by inserting a glowing match stick or incense stick which would burst into flames. This shows the presence of oxygen.

- What precautions do you need while removing test tube from the beaker. Discuss with your teacher.

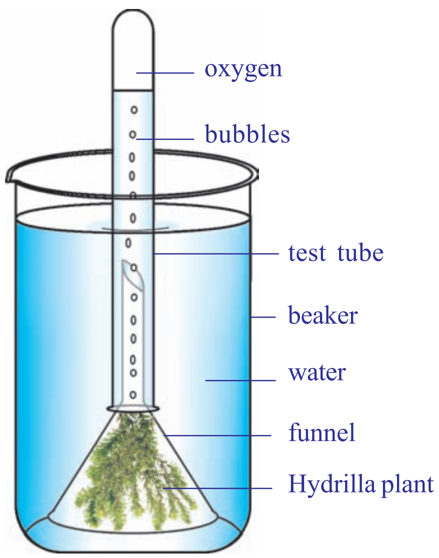


Fig-5: Hydrilla experiment

Activity-3

Sunlight is necessary to form starch in green plants

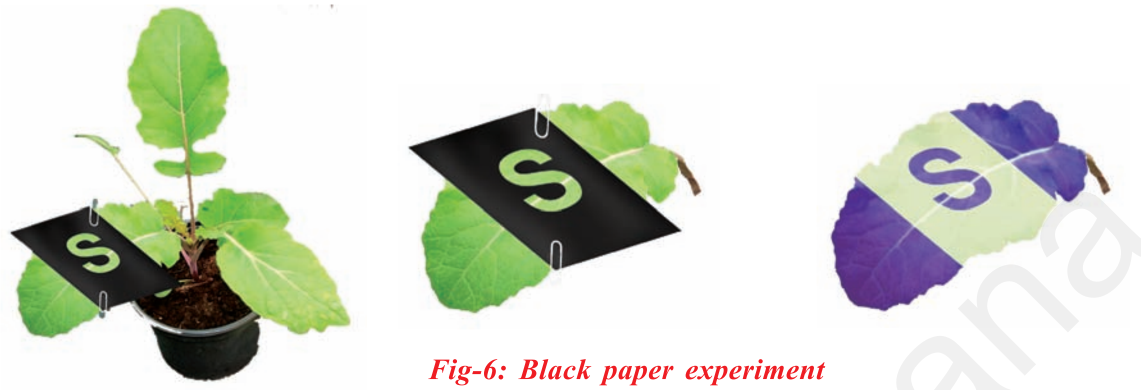


Fig-6: Black paper experiment

- Take a potted plant with destarched leaves. Remember the process of destarching leaves mentioned in activity-1.
- Cover one of its leaves with black paper on which a design is cut. Fix the paper on the leaf in such a manner that light does not enter the dark part.
- Place this potted plant in sun light.
- After few hours of exposure to bright sunlight, test the leaf which is covered by black paper for the presence of starch.
- *Which part of the leaf turns bluish-black? What about the remaining part?*
- *Observe the colour of leaf stained with iodine. Can you tell why it is stained differently?*
- It will be observed that only the parts of the leaf, which could get light through the cut out design, turns bluish-black showing the presence of starch.

Chlorophyll and Photosynthesis

Ingenhousz wanted to find out more about photosynthesis and carried out several other experiments. He proposed that only green plant parts could carry out the process of photosynthesis.

What about plants having coloured leaves? How is it that new leaves which look dark red in colour in several plants turn green? Do plants having reddish or yellowish leaves also carry out photosynthesis? What made plants carry out photosynthesis while even green coloured animals (like some birds)could not? Questions like these remained challenges until scientists could isolate the green coloured substance from plant parts and study its nature.

Establishment of Ingenhousz's proposition become possible in the year 1817 due to the work of two scientists Pelletier and Caventou. They obtained an extract of the green coloured substance and named it 'chlorophyll' (meaning green leaf).

It was also found that pigments in addition to chlorophyll, carotenoids and phycobilins could also aid in the process of photosynthesis by passing on the energy of sunlight trapped by them to chlorophyll.

Where does Photosynthesis take place?

- *Try to name some parts where you think photosynthesis occurs.*
- *Where is chlorophyll and other pigments present in the plant?*
- *Do you think the new reddish leaves of plants also carry out photosynthesis?*

The exact location of the photosynthetic part or a part containing chlorophyll was not known till another six decades after the discovery of chlorophyll. In 1883, Julius Von Sachs, observed that chlorophyll is found in organelles within the cell. Such organelles were named as 'chloroplasts'. These are present in large numbers in the cells (around 40 – 100) of parts like the stomatal guard cells and ground tissues of green parts of the plant.

You have studied about Chloroplast in Class IX. Let us observe the figure of TS of leaf showing Chloroplast in Palisade and Spongy Parenchyma (ground tissue).

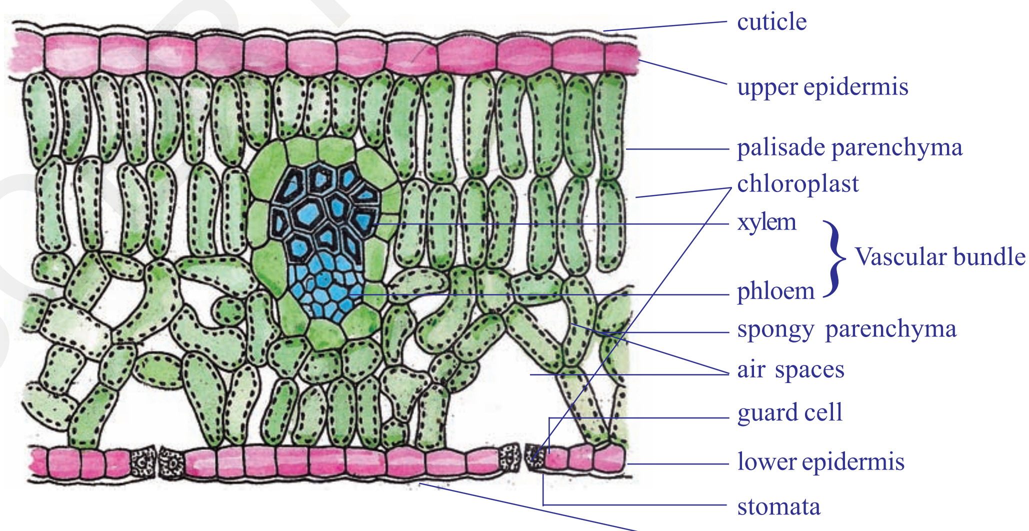


Fig-7: T-S of leaf

Do you know?

If a cell is broken up, the chloroplasts also break into pieces, so it becomes a very difficult task to isolate them to study the different steps of photosynthesis.It was not until 1954, that Daniel I. Arnon was able to break up plant cells so gently that whole chloroplasts could be obtained for study in the lab.

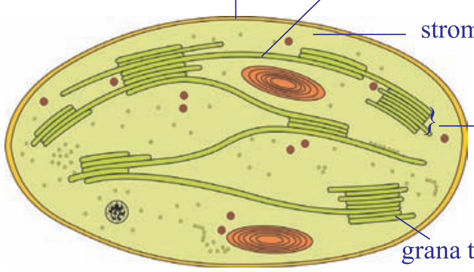


Fig-8: T-S of chloroplast

It has been found that the chloroplast is a membranous structure, consisting of 3 membranes. The inner most layer forms stacked sack like structures called as grana. It is believed to be a site for trapping of solar energy. The intermediary fluid filled portion is called as stroma. It is believed to be responsible for enzymatic reactions leading to the synthesis of glucose, which in turn join together to form starch.

Substances found in chloroplast which capture sunlight are called photosynthetic pigments. There are several types of photosynthetic pigments involved in the process to produce organic molecules like glucose in plants.

Chlorophyll is such a pigment which contain one atom of magnesium. It is similar in structure to the haem of haemoglobin (The iron containing red pigment that transports oxygen in blood.). Two major kinds of chlorophylls are associated with thylakoid membranes. Chlorophyll 'a' is bluish-green in colour and chlorophyll 'b' is yellowish-green colour. Around 250 to 400 pigment molecules are grouped as light harvesting complex or photosynthetic unit in each thylakoid. Such innumerable units function together in chloroplasts of green plants in the process of photosynthesis.

During photosynthesis several events occur in the chloroplast some of them are:

1. Conversion of light energy to chemical energy
2. Splitting of water molecule (photolysis of water)
3. Reduction of carbondioxide to carbohydrates

Light is required to initiate several events while several may continue even in absence of it. That would mean, once light energy has been captured it can help reactions to continue even in the dark. Light dependent events or reactions are called light reactions and it has been found to take place in grana, while the rest are called light independent or dark reactions and they occur in the stroma.

Mechanism of Photosynthesis

1. Light dependent reaction

In this reaction light plays a key role. A series of chemical reactions occur in a very quick succession initiated by light and therefore the phase is technically called the photochemical phase or light dependent reaction. The light reaction takes place in chlorophyll containing thylakoids called grana of chloroplasts. Several steps occur in the light dependent reaction.

Step-I : The chlorophyll on exposure to light energy becomes activated by absorbing photons. (Photon is a unit of light energy)

Step-II: The energy is used in splitting water molecule to release O_2. The reaction is known as photolysis (photo means light, lysis means breaking). This was discovered by Robert Hill. Hence it is also called Hill's reaction.

Hydrogen produced in photolysis is immediately picked up by special compound NADP (Nicotinamide Adenine Dinucleotide Phosphate) to form NADPH (Reduced NADP). Another energy rich compound ATP (Adenosine triphosphate) is also formed at the end of the light reaction.

2. Light independent reaction

This reaction does not require the presence of light and extension of the phases after day time may occur in some plants. This is also called dark reaction. But the term dark reaction does not mean that it occurs when it is dark at night. It only means that these reactions do not depend on light. In the dark phase the hydrogen combines with CO_2 by utilizing ATP energy and produces glucose ($C_6H_{12}O_6$). This synthesis occurs in a number of steps using certain special intermediate compounds and enzymes. Finally glucose may be converted into storage product starch.

Plants are capable of surviving under a range of situations, from very hot, dry and brightly lighted conditions to wet, humid and dimly lighted condition. The requirement of light and other factors varies from one plant to another.

Heterotrophic Nutrition

In the living world all organisms are capable of surviving under different conditions and acquring their food in different ways. We have studied about organisms that can capture light to produce their food. These are autotrophic in nature. While those that can not are heterotrophic.

How do organisms obtain their nutrition

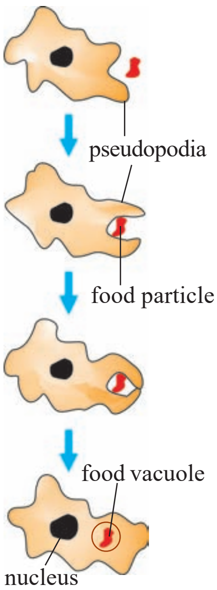


Fig-9(a): Nutrition in Ameoba

Depending on the type and availability of food organisms can adapt to a range of strategies of food intake and use. Some organisms break down the food materials outside the body and then absorb it. For example, mold, yeast, mushrooms etc. which are called saprophytes. Some other organisms derive nutrition from plants or animals without killing them. This type of parasitic nutritive strategy is used by a wide variety of organisms like Cuscuta, lice, leeches and tape worms. Others take in whole material and break it down inside their bodies. What can be taken in and broken down depends on the bodys' design and it's function.

Since the food and the way it is obtained differs, the digestive system is also different in various organisms. In single celled organisms, like amoeba the food may be taken in by the entire surface but as the complexity of the organism increases, different parts become specialized to perform different functions.

For example amoeba [fig-9(a)] takes in food using temporary finger like extensions (pseudopodia) of the cell surface which fuse over the food particle forming food vacuole. Inside the food vacuole, complex substances are broken down into simpler ones. Then diffuse into the cytoplasm. The remaining undigested material is moved to the surface of the cell and thrown out. In Paramoecium [fig-9(b)], which is also a unicellular organism the cell has a definite shape. Food is taken in at a specific spot. Food is moved to the spot by the movement of cilia which covers the entire surface of the cell, where the food is ingested (cytostome).

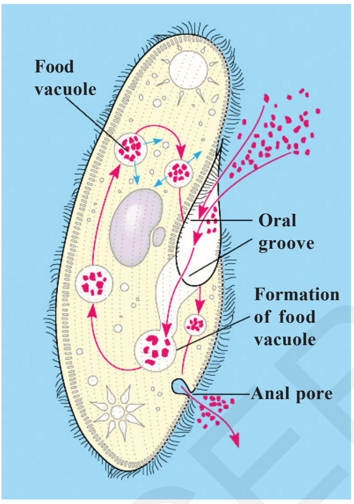


Fig-9(b): Nutrition in Paramoecium

Parasitic nutrition in Cuscuta

Dodder (genus : *Cuscuta*) is a leafless, twining, parasitic plant belongs to morning glory family (Convolvulaceae). The genus contains about 170 twining species that are widely distributed throughout the temperate and tropical regions of the world.

The dodder plant contains no chlorophyll. *Cuscuta reflexa* has been found to have very little amount of chlorophyll hence it absorbs food through haustoria. Haustoria are root like structures that penetrate the tissue of a host plant and may kill it. The slender, string like stems of the dodder may be yellow, orange, pink, or brown in colour. Its leaves are reduced to minute scales. Its flowers, in nodule like clusters, are made up of tiny yellow or white bell-like petals.

The dodder's seed germinates, forming an anchoring root, and then sends up a slender stem that grows in a spiral fashion until it reaches a host plant. It then twines around the stem of the host plant and forms haustoria, which penetrate through it. Water is drawn through the haustoria from the host plant's xylem, and nutrients are drawn from its phloem.

Fig-10: Haustoria in cuscuta

Nutrition in Human Beings

Human digestive system is very complex in nature. Different parts are involved and perform different functions by using various digestive juices and enzymes.

Let us observe the figure of digestive system.

The alimentary canal is basically a long tube extending from the mouth to the anus. We can see that this tube has different parts. Various regions are specialized to perform different functions.

- *What happens to the food once it enters our body?*

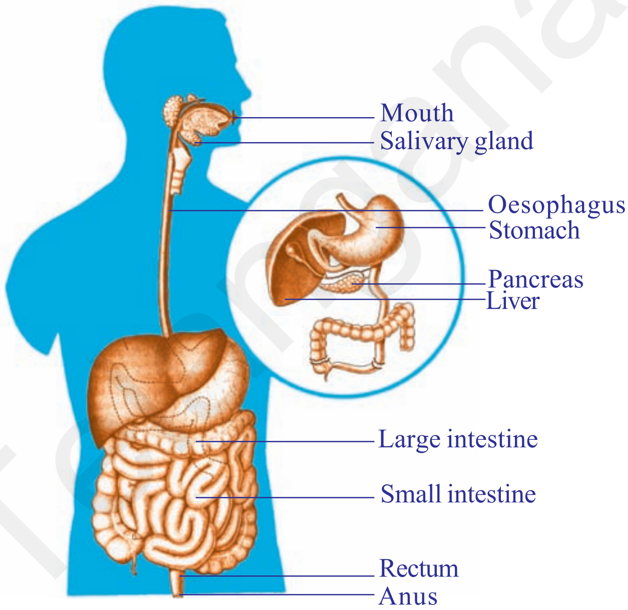


Fig-11: Digestive system

We eat various types of food which pass through the same digestive tract. It is to be converted to substances small enough to be utilised by our body. This needs various processess that can be studied as follows.

Passage of food through alimentary canal or gut

Food is cut and crushed by our teeth in the mouth and mixed with saliva to make it wet and slippery lump called **bolus** (This process is called as mastication). This bolus is suitable to pass through oesophagus. Saliva is secreted by three pairs of salivary glands. Saliva mainly contains an enzyme amylase (ptyalin) which helps in the breakdown of complex carbohydrates to simple ones. This process of breaking down of complex substances into simple substances with the help of enzymes and their absorption into the body is

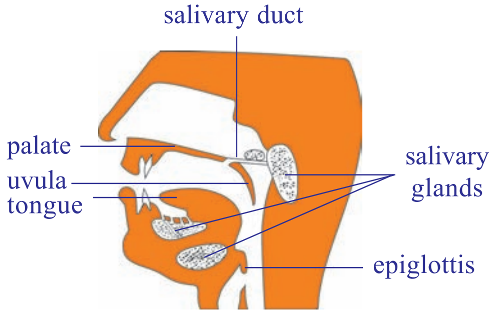


Fig-12: Location of some salivary glands called digestion. The tongue helps in mixing the food and pushing it into the next part. The lower jaw also helps in this process.

We can find out the effect of salivary amylase on carbohydrates to observe what might be happening in our mouth.

To know the chemical nature of the saliva perform the activity-7 of co-ordination in life process of this text book.

The soft food mixed with saliva passes through oesophagus or food pipe by wave like movements called peristaltic movement to the stomach.

At the stomach, food gets churned with gastric juice and HCl rendering the medium acidic. Now the food is in semisolid condition. The digestion of food goes on as most of the proteins are broken down into smaller molecules with the help of enzyme pepsin.

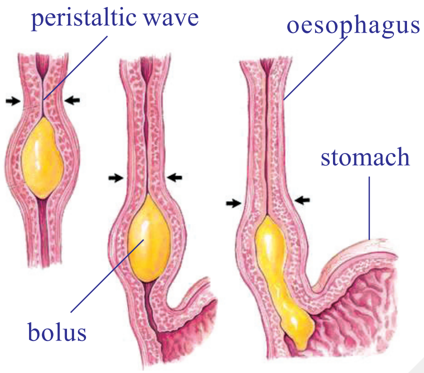


Fig-13: Peristaltic movement

Food in the form of a soft slimy substance where some proteins and carbohydrates have already been broken down is called chyme. Now the food material passes from the stomach to the small intestine. Here the ring like muscles called pyloric sphincters relax to open the passage into the small intestine. The sphincters are responsible for regulating the opening of the passage such that only small quantities of the food material may be passed into the small intestine at a time.

The small intestine is the longest part of the alimentary canal. Its proximal part is called **duodenum**. It is the site of further digestion of carbohydrates, proteins and fats. It receives the secretion of liver and pancreas for this purpose. These juices render the internal condition of the intestine gradually to a basic or alkaline one.

Fats are digested by converting them into small globule like forms by the help of the bile juice secreted from liver. This process is called emulsification.

Pancreatic juice secreted from pancreas contains enzymes like trypsin for carrying on the process of digestion of proteins and lipase for fats.

Walls of the small intestine secrete intestinal juice which carry this process further that is small molecules of proteins are broken down to further smaller molecules. The same is the condition with fats. Carbohydrate digestion that started in the mouth and did not occur in the stomach, resumes now as the medium gradually changes to an alkaline one and the enzymes become active for carbohydrate breakdown.

Activity-4

Studying the enzymes chart

Let us study the chart showing different enzymes and digestive juices and their functions.

Table-1: Digestive enzymes

S.No.	Enzyme/Substance	Secreted by	Secreted into	Digestive juice	Acts on	Products
1	Ptyalin (salivary amylase)	Salivary glands	Buccal cavity	Saliva	Carbohydrates	Maltose
2	Pepsin	Gastric glands	Stomach	Gastric juice	Proteins	Peptones
3	Bile (No enzymes)	Liver	Duodenum	Bile juice	Fats	Emulsification (breaking down of large fats into small globules)
4	Amylase	Pancreas	Duodenum	Pancreatic juice	Carbohydrates	Maltose
5	Trypsin	Pancreas	Duodenum	Pancreatic juice	Proteins	Peptones
6	Lipase	Pancreas	Duodenum	Pancreatic juice	Fats	Fatty acids and glycerol
7	Peptidases	Intestinal glands	Small Intestine	Intestinal juice	Peptides	Amino acids
8	Sucrase	Intestinal glands	Small Intestine	Intestinal juice	Sucrose (Cane Sugar)	Glucose

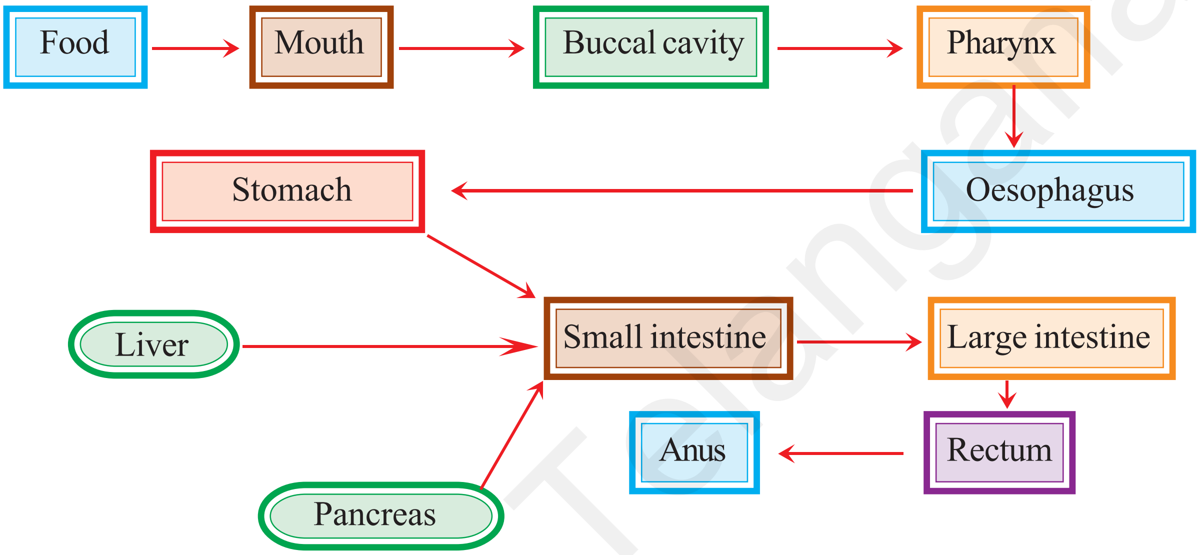
- *Name the enzymes which act on carbohydrates?*
- *Which juice contains no enzymes?*
- *What are the end products of fats?*
- *What are the enzymes that act on proteins?*

Transport of the products of digestion from the intestine into blood (through the wall of intestine) is called absorption. Internally, intestinal wall has a number of finger like projections called Microvilli. The villi increase the surface area for absorption. Blood vessels and lymph vessels are present in the form of a network in the villi.

Products of digestion are absorbed first into the villi and from here into the blood vessels and lymph vessels. Thus after maximum absorption of food in the small intestine the rest passes into the large intestine. Here most of the water present is taken up from this material. This material is then expelled through the anus which is the last part of the alimentary canal. This passage of undigested material from the body by the way of anus is called defecation. Food that passes out of the anus still contains considerable amount of proteins, fats and carbohydrates, roughages or fibres of either carbohydrates or proteins. We will learn some more points about the coordination of digestive system with other systems in the chapter coordination in life processes.

Flow chart of human digestive system

- *What do you think about the necessity of the process of digestion?*
- *What are its major steps?*



Health aspects of the alimentary canal

The human alimentary canal usually functions remarkably well considering how badly we treat it on occasions! Sometimes it rebels, and we either feel sick or have indigestion.

Vomiting is the body's method of ridding itself of unwanted or harmful substances from the stomach. The peristaltic movements of the stomach and oesophagus reverse their normal direction and the food is expelled. There are many causes of vomiting, but one of the most common is over eating, especially when the food contains a high proportion of fat. Vomiting also occurs when we eat something very indigestible or poisonous.

When we have a greenish vomit usually called as 'bilious' or 'liverish', we get a bitter taste and it is often the result of having over eaten. The liver is unable to cope with the excessive fat and we get a feeling of nausea.

Indigestion is a general term used when there is difficulty in digesting food. Healthy people can usually avoid problems related to digestion by:

a) having simple, well balanced meals

b) avoiding taking violent exercise soon after eating food

c) Drinking plenty of water.

d) Intake of fibre rich food to avoid constipation.

A serious condition of indigestion is caused by stomach and duodenal ulcers. The cause of this may be the kind of diet, an infection or habit. You have studied about recent researches in the peptic ulcers caused by some bacteria in class-VIII.

Diseases due to malnutrition

Food helps to maintain life processes. Our diet should be a balanced one which contains proper amount of carbohydrates, proteins, vitamins, mineral salts and fats. Two third of world population is affected by food related diseases. Most of them are facing various diseases due to lack of balanced diet. It is very important to discuss about food deficiency diseases.

Eating of food that does not have one or more than one nutrients in required amount is known as malnutrition. Poor health, willful starvation, lack of awareness of nutritional habits, socio-economic factors are all the reasons for malnutrition in our country.

Malnutrition may be-

1. Protein malnutrition,
2. Calorie malnutrition,
3. Protein calorie malnutrition.

Let us observe harmful effects of malnutrition in children.

Fig-14: Kwashiorkor

1. Kwashiorkor disease: This is due to protein deficiency in diet. Body parts become swollen due to accumulation of water in the intercellular spaces. Very poor muscle development, swollen legs, fluffy face difficult to eat, diarrhoea, dry skin are the symptoms of this disease.

Fig-15: Marasmus

2. Marasmus: This is due to deficiency of both proteins and calories. Generally this disease occurs when there is an immediate second pregnancy or repeated child births. Lean and weak, less developed muscles, dry skin, diarrhoea, etc., are the symptoms of this disease.


Fig-16: Obesity

Obesity: Apart from the deficiency diseases mentioned above, the constant intake of high calorific food may lead to a condition called obesity. In this condition a person gains extra fat and becomes overweight. Obese children may suffer from problems related to cardiovascular, renal, gallbladder etc.


Fig-17: Pellagra

Vitamin deficiency diseases

Vitamins are organic substances. They are micro nutrients required in small quantities. Actually vitamins are not synthesised in the body, we do not generally suffer from vitamin deficiency. The source of vitamins to our body is through two ways. One is diet and other is bacteria present in the intestine that synthesises and supplies vitamins to the body.

Vitamins are classified into two groups. One is Water soluble vitamins (B-complex, vitamin C) and other is fat soluble vitamins (vitamin A, D, E and K). If food we take doesn't contain vitamins in right proportion, it can lead to vitamin deficiency disorders. Let us study the following chart showing vitamins available sources and deficiency diseases.

Table-2: Vitamins

Vitamin	Resources	Deficiency diseases	Symptoms
Thiamin (B1)	Cereals, oil seeds, vegetables, milk, meat, fish, eggs.	Beri beri	Vomitings, fits, loss of appetite, difficulty in breathing, paralysis.
Riboflavin (B_2)	Milk, eggs, liver, kidney, green leafy vegetables.	Glossitis	Mouth cracks at corners, red and sore tongue, photophobia, scaly skin.
Niacin (B_3)	Kidney, liver, meat, egg, fish, oil seeds.	Pellagra	Dermatitis, diarrhoea, loss of memory, scaly skin.
Pyridoxine (B_6)	Cereals, oil seeds, vegetables, milk, meat, fish, eggs, liver.	Anaemia	Hyper irritability, nausea, vomiting, fits.
Cyanocobalamin (B_{12})	Synthesised by bacteria present in the intestine.	Pernicious anaemia	Lean and weak, less appetite.
Folic acid	Liver, meat, eggs, milk, fruits, cereals, leafy vegetables.	Anaemia	Diarrhoea, loss of leucocytes, problems related to mucus in the intestines.
Pantothenic acid	Sweet potatoes, ground nuts, vegetables, liver, kidney, egg.	Burning feet	Walking problems, sprain.
Biotin	Pulses, nuts, vegetables, liver, milk, kidney.	Nerves disorders	Fatigue, mental depression, muscle pains.
Ascorbic acid (C)	Green leafy vegetables, citrus fruits, sprouts.	Scurvy	Delay in healing of wounds, fractures in bones.
Retinol (A)	Leafy vegetables, carrot, Tomato, pumpkin, papaya, mango, meat, fish, egg, liver, milk, cod liver oil, shark liver oil.	Eye, skin diseases	Night blindness, xerophthalmia, cornea failure, scaly skin.
Calciferol (D) (sunshine vitamin)	Liver, egg, butter, cod liver oil, shark liver oil, sun rays stimulate the formation of vitamin D from the sub-cutaneous fat.	Rickets	Improper formation of bones, Knock-knees, swollen wrists, delayed dentition, weak bones.
Tocoferol (E)	Fruits, vegetables, sprouts, sunflower oil.	Fertility related disorders	Sterility in males, abortions in females.
Phylloquinone (K)	Green leafy vegatables, milk, meat, egg.	problems related to Blood clotting	Delay in blood clotting, over bleeding.

Key words

Glucose, starch, cellulose, chloroplast, grana, stroma, light reaction, dark reaction, heterotrophic nutrition, parasitic nutrition, haustoria, Alimentary canal, salivary glands, peristaltic movement, amylase, ptyalin, pepsin, chyme, sphincter, digestion, pancreas, enzymes, villi, bile juice, lipase, fat, liver, emulsification, kwashiorkor, marasmus.

What we have learnt

- Autotrophic nutrition involves the intake of simple inorganic materials like some minerals, water etc. from the soil. Also some gases from the air. Carbohydrates are synthesized with the help of energy from the sun.
- Photosynthesis is the process by which living plant cells containing chlorophyll, produce food substances [glucose & starch] from carbon dioxide and water by using light energy. Plants release oxygen as a waste product during photosynthesis.
- Photosynthesis process can be represented as

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{light}} C_6H_{12}O_6 + 6H_2O + 6O_2$$

- The materials required for photosynthesis are light: Carbon dioxide, Water, photosynthetic pigment chlorophyll.
- Chloroplasts are the sites of photosynthesis. Light reaction takes place in the grana region and light independent reaction takes place in the stroma region of chloroplast.
- The end products of photosynthesis are glucose, water and oxygen.
- During photosynthesis the important events which occurs in the chloroplast are-
  a) Conversion of light energy into chemical energy
  b) Splitting of water molecule
  c) Reduction of carbon dioxide to carbohydrates
- Heterotrophic Nutrition involves the intake of complex material prepared by other organisms.
- The form of nutrition differs depending on the type and availability of food material as well as how it is obtained by the organism.
- In single celled organisms the food may be taken in by the entire surface but as the complexity of the organism increases different parts becomes specialized to perform different functions.
- The large complex food molecules such as carbohydrates, proteins , lipids, etc., are broken down in to simple molecules before they are absorbed and utilized by the animals. This process of breaking down of complex molecules into simple molecule is called digestion.
- In human beings the food eaten is broken down in various steps with the help of enzymes secreted by digestive glands which are associated with the alimentary canal and the digested food is absorbed in small intestine to be sent to all cells in the body.

- The digestive system includes the alimentary tract and several associated organs and digestive glands. The functions of system are as follows :
  - a) Ingestion: Taking of food into the body
  - b) Digestion: Breaking up of complex food substances into the simple substances by specific enzymes. So that they can be used by the body.
  - c) Absorption: The passage of digested food through the walls of alimentary tract (particularly in small intestine) into circulatory system.
  - d) Defecation: The passage of undigested material from the body by the way of anus.

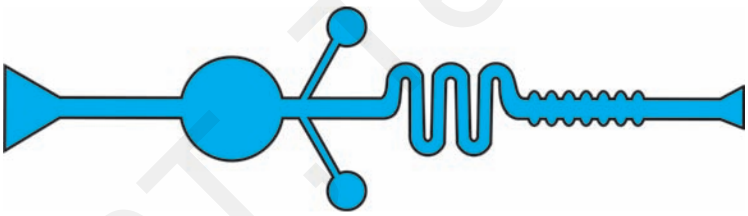
Improve your learning

1. Write differences between (AS1)
   a) autotrophic nutrition - heterotrophic nutrition
   b) Ingestion - digestion c) Light reaction - dark reaction d) Chlorophyll - chloroplast
2. Give reasons (AS1)
   a) Why photosynthesis is considered as the basic energy source for most of living world?
   b) Why is it right to call the dark phase of photosynthesis as a light independent phase?
   c) Why is it necessary to destarch a plant before performing any experiment on photosynthesis?
   d) Why is not possible to demonstrate respiration in green plant kept in sun light?
3. Give examples (AS1)
   a) Digestive enzymes b) Organisms exhibiting heterotrophic nutrition
   c) Vitamins d) Nutritional deficiency diseases
4. From where do plants get the raw materials required for photosynthesis?(AS1)
5. Explain the process of photosynthesis in with the help of a flow chart? (AS1)
6. Name three end products of photosynthesis? (AS1)
7. What is the substance that connects light reaction and dark reaction? (AS1)
8. In most of leaves the upper surface will be more green and shiny than the lower surface. Why? (AS1)
9. Explain the structure of chloroplast with a neatly labeled sketch. (AS5)
10. What is the role of acid in stomach? (AS1)
11. Mention the names of glands and organs which help in digestion? (AS1)
12. How is the small intestine designed to assimilate the food? explain. (AS1)
13. How are fats digested? Where do they get digested? (AS1)
14. What is the role of saliva in the digestion of food? (AS1)
15. What will happen to protein digestion as the medium of intestine is gradually rendered alkaline? (AS1)
16. What is the role of roughages in the alimentary tract? (AS1)
17. What is malnutrition? Explain few nutritional deficiency diseases. (AS1)
18. How do fungi and bacteria obtain their nourishment? (AS2)

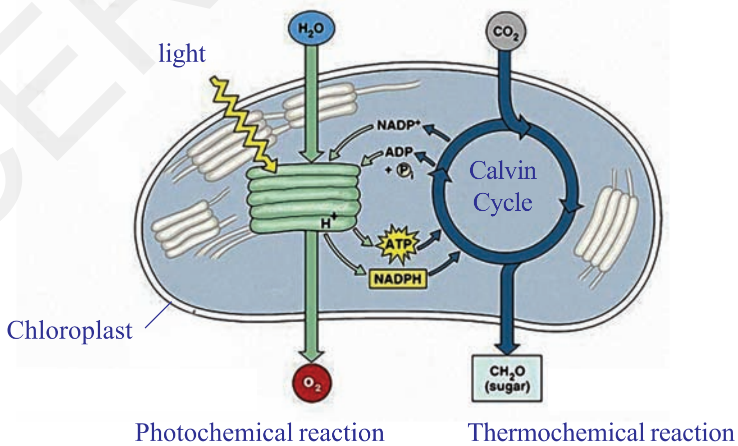
19. If we keep on increasing CO_2 concentration in air what will happen to the rate of photosynthesis?(AS2)
20. What happens if the rate of respiration is more than the rate of photosynthesis in a plant?(AS2)
21. How can you say that the carbohydrates are not completely digested until they reach the intestine?(AS1)
22. What process do you follow in the laboratory to study the presence of starch in leaves?(AS3)
23. How would you demonstrate that green plant releases oxygen when exposed to light?(AS3)
24. Visit a primary health centre and collect the information about the children at diffrent ages suffering from malnutrition. (AS4)

S.No.	Age group	No. of children with malnutrition		
		Protiens	Calories	Vitamins
1.				
2.				

25. Would the survival of organisums become difficult, if there are no green plants on the earth? Why/ Why not? (AS1)
26. Draw the labelled diagram of human digestive system? List out the parts where peristalsis takes place. (AS5)
27. Raheem prepared a model showing the passage of the food through different parts of the alimentary canal? Observe this and label it's parts. (AS5)



28. Observe the following diagram and write a note on light dependent, light independent reactions.(AS5)



29. What facts about the green plants do you appreciate?(AS6)
30. What food habits will you follow after reading this chapter? Why? (AS7)

Fill in the blanks

1. The food synthesized by the plant is stored as ______________________.
2. ________________________ are the sites of photosynthesis.
3. The enzymes in the pancreatic juice help in the digestion of ___________________ and ________________________.
4. The finger like projections which increases the surface area in small intestine are called___________________.
5. The gastric juice contains _________________________ acid.
6. ___________________ vitamin is synthesised by bacteria present in intestine.

Choose the correct answer

1. Which of the following organisms take the food by parasitic nutrition? ( )
   (i) Yeast (ii) Mushrooms (iii) Cuscuta (iv) Leeches
   A) (i), (ii) B) (iii) C) (iii), (iv) D) (i)
2. The rate of Photosynthesis is not affected by: ( )
   A) Light Intensity B) Humidity C) Temperature D) Carbon dioxide concentration
3. A plant is kept in dark for about forty eight hours before conducting any experiment on Photosynthesis in order to : ( )
   A) Remove chlorophyll from leaves
   B) Remove water from leaves
   C) Ensure that no photosynthesis occurred
   D) Ensure that leaves are free from the starch
4. The digestive juice without enzyme is ( )
   A) Bile B) Gastric juice C) Pancreatic juice D) saliva
5. In single celled animals the food is taken by ( )
   A) body surface B) Mouth C) Teeth D) Vacuoles
6. Which part of the plant takes in carbondioxide from the air for photosynthesis ( )
   A) Root hair B) Stomata C) Leaf veins D) Sepals

Chapter

2

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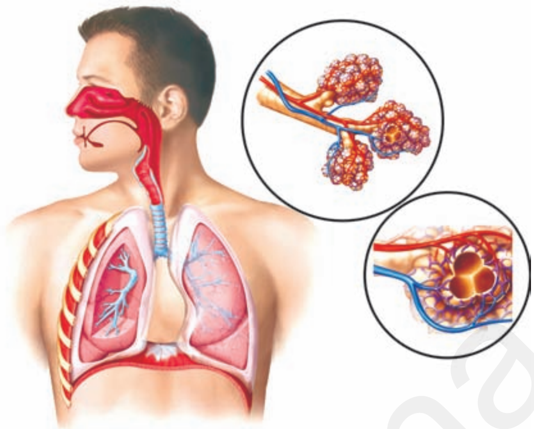


Respiration

Using food to carry out life processes is key to life for all living beings in both multicellular or unicellular. In the chapter, on nutrition we have discussed how the body draws out nutrients from the food taken in. The food provides energy for all the biological activities only after break down through the process known as respiration. Thus, respiration leads to final utilization of food. Normally respiration takes place in oxygen rich environment. Cells of the living body use food constantly to help our body to function properly. They require the presence of air, food material and some chemicals.

The term 'respiration' derived from Latin word 'respire' meaning 'to breathe', refers to the whole chain of processes from the inhalation of air to the use of oxygen in the cells. To begin with, we shall study the relation of gases and the process of respiration.

Discovery of gases and respiration

The term respiration came into use, a century after the word breathing was used, way back in the 14^{th} century. It was used much before people knew that air is a mixture of gases. They hardly knew anything about all the life processes that took place internally in a living body. Respiration which was used as a medical term, usually referred to as a process involving passage of air and production of body heat.

Fig-1: Lavoisier

It was not until 18^{th} century when Lavoisier and Priestley did a comprehensive work on properties of gases, their exchange and respiration that we came to know something about how the process of gaseous exchange goes on, in our body. You have already studied about some of Priestley's experiments in earlier classes (You have an account of it in the chapter on nutrition as well). Recall the concepts and answer the following.

- *Can it be said that Priestley's experiment helped us to find out more about composition of air? How?*

Lavoisier also carried out several experiments to understand the properties of gases. In his early experiments, it is clear that Lavoisier thought that the gas liberated on heating powdered charcoal in a bell jar kept over water in a trough was like fixed air. In those days carbon dioxide was known as fixed air. The next series of experiments dealt with the combustion of phosphorus in a bell jar. From these studies Lavoisier showed that whatever in the atmospheric air which combined with the phosphorus, was not water vapour. His final words are that the substance which combines with the phosphorus is "either air itself, or another elastic fluid present in a certain proportion, in the air which we breathe". This was the respirable air, a component of air that also helped in burning.

- *What was produced by combustion according to Lavoisier?*
- *What did Lavoisier find out about air from his experiments?*
- *What conclusion can be drawn from Lavoisier's experiments?*


Fig-2: Priestley

The air that we breathe out made lime water milky while that after heating metal did not. He noted that fixed air also made lime water milky.

He drew immediately the logical conclusion regarding the process of respiration. Either eminently respirable air is changed in the lungs to fixed air; or an exchange takes place, the eminently respirable air being absorbed, and an almost equal volume of fixed air being given up to the air from the lungs.

Lavoisier's findings lead way to several other researchers.

- *Which gas do you think is Lavoisier talking about when he says fixed air?*
- *Which gas according to him is respirable air?*
- *What steps in the process of respiration does Lavoisier mention as an inference of his experiments?*

A few lines from a textbook of Human Physiology, written by a renowned chemist, John Daper around mid-19th century goes like this..

"The chief materials which a living being receives are matter that can be burnt, water and oxygen gas; and out of the action of these upon one another, all the physical phenomena of its life arise. What the body expels out is water, oxide of carbon, phosphorous, sulphur and others."

Thus, we can see that the role of major compounds and elements in the process of respiration was known by mid 19th century. The events involved were not very clearly understood, but people believed that there was some relationship between the heat produced in the body and the process of respiration.

- *It is a common observation that our breath is warmer than the air around us; does respiration have anything to do with this?*

Let us study the steps involved in respiration in human beings to figure it out.

Steps in Respiration

There are no strict demarcations of steps involved in the process of respiration. It is a very complex process of several physical and biochemical processes. But for a general understanding on what goes on, we shall study under the following heads.

Breathing →	Gaseous exchange at lungs level →	Gas transport by blood →	Gaseous exchange at tissue level →	Cellular Respiration
Air movement into and out of lungs	Exchange of gases between alveoli and blood	Transport of oxygen from blood capillaries of alveoli to body cells and return of carbon dioxide	Exchanging of oxygen from blood into the cells and carbon dioxide from cells into the blood	Using oxygen in cell processes to produce carbon dioxide and water, releasing energy to be used for life processes

Breathing

In the previous classes we did experiments to find out what was there in the air we breathe out.

We had seen that in a set up with lime water, it turned milky white fast as we breathe out into it as compared to a similar set up in which normal air was passed with the help of a syringe or pichkari in lime water. (Experimental set up to test the presence of Carbondioxide in exhaled air). Arrange apparatus as shown in figure-3 and try to do the experiment once again to find out what happens.

- *What does this experiment indicate?*
- *Which gas turns lime water milky?*
- *Which gas do you think might be present in greater quantities in the air we breathe out as compared to air around us?*

- *We are also aware of the fact that water vapour deposits on a mirror if we breathe out on it.*
- *Where does this water vapour come from in exhaled air?*

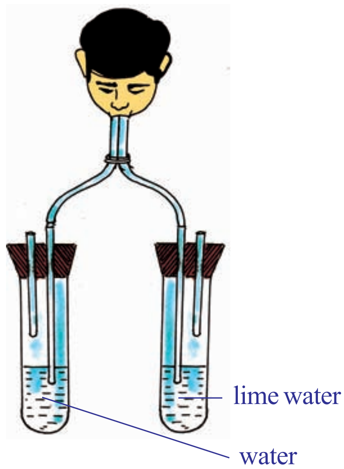


Fig-3: Presence of CO_2

We shall have to study the pathway of air in our body through our respiratory system and the mechanism of breathing in respiration to find out how the exhaled air comes out. By "respiratory system" we usually mean the passages that transport air to the lungs and to the microscopic air sacs in them, called alveoli (where gases are exchanged between them and blood vessels) and vice versa.

Pathway of air

Let us observe the pathway of air from nostril to alveolus.

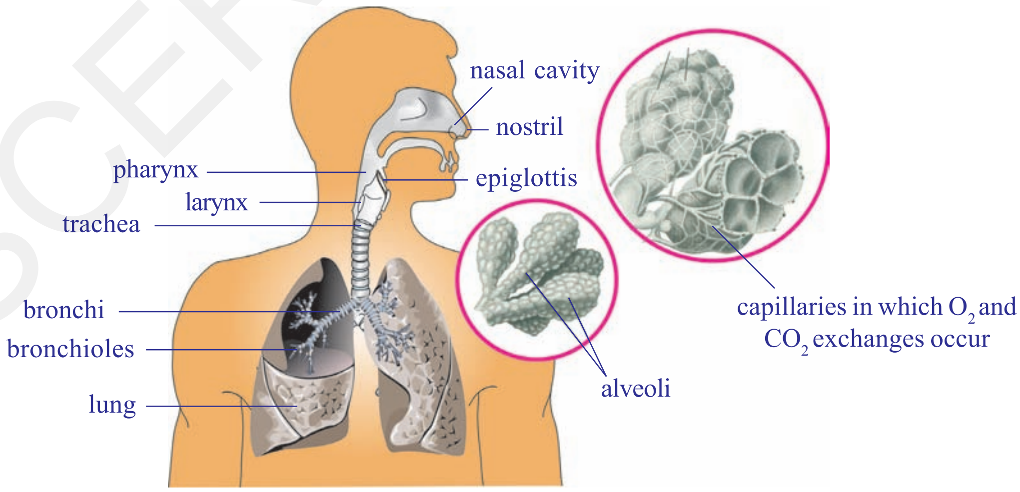


Fig-4: Respiratory system of man

Nostrils: Air usually enters the body through the nostrils

Nasal cavity: Air is filtered. The moist surface of the lining of the nasal cavity, and the hairs growing from its sides, remove some of the tiny particles of dirt in the air. In addition, as the inhaled air passes through the nasal cavity, its temperature is brought close to that of the body, and it takes up water vapour so that it becomes more moist than before.

Pharynx: Warming and moistening goes on in this common passage of digestive and respiratory system. Epiglottis, a flap like muscular valve controls movement of food and air towards their respective passages.

Larynx: This stiff box contains our vocal cords. When air passes out of the lungs and over the vocal cords, it causes them to vibrate. This produces sounds on the basis of our speech, song etc.

Trachea: Wind pipe channeling air to lungs. Touch your throat to feel the tube like structure.

Bronchus: At its lower end the trachea or the wind pipe divides into two bronchi-one leading to each lung.

Bronchioles: Each bronchi is further divided into smaller and smaller branches called bronchioles.

Alveolus: These finally terminate in clusters of air sacs called alveoli in the lungs which are very small and numerous. Gaseous exchange takes place here as blood capillaries take up oxygen and expel carbon dioxide.

Blood: Blood carries oxygen, to each and every cell of the body and collects CO_2 from them.

The whole passage from nostrils to alveolus is moist and warm.

Do you know?

The interior of lung is divided into millions of small chambers, thus tremendously increasing the moist surface available for transfer of gases between air and blood. The linings of the lungs are much folded and so their total surface is enormous. If all alveoli of our lungs are spread out they will cover an area of nearly 160 m^2. Which is equivalent to tennis court.

Think and discuss

- What will happen if the respiratory tract is not moist?
- Are both lungs similar in size?
- Why are alveoli so small and uncountable in number?

Epiglottis and passage of air

From the nasal cavity, the air goes into the pharynx. There is a tricky problem here. From the pharynx there are two passages, beginning with nearly same opening and ending into separate ones, one to the lungs and one to the stomach. It is important that air goes into one and food into the other. It is also important that food *does not* enter the tube through which air goes into the lungs. The movement of food and air is channeled properly by a flap like valve, the ***epiglottis*** that protects the tube to the lungs, prohibiting entry of food. Observe the Fig-5(a), Fig-5(b) and discuss in your class how epiglottis works while breathing or swallowing.

Epiglottis diverts air to lungs

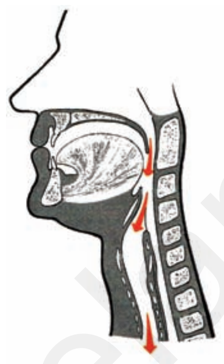


Fig-5(a): Breathing

Epiglottis diverts food mass away from opening of larynx

Fig-5(b): Swallowing

This valve is partly closed when we swallow food; it deflects food down to the stomach and keeps it out of the trachea or wind pipe which is the route to the lungs. The epiglottis opens more widely when we take a breath, and air enters the lungs. Nervous regulation is important in guiding the function of epiglottis and passage of food and air. Let us try to do an activity to feel what happens when we swallow food.

- *Why are we advised not to talk while eating food ?*

Activity-1

Keep your palm around an inch away from your nose, feel you are breathing out, do not remove it until you have finished the activity. Breathe steadily for 1-2 minutes. Now take a piece of any fruit, chew and before swallowing it keep the fingers of the other palm on your throat, now swallow it.

- *What did you notice? What happens to your breath as you try to swallow?*

- *What is helping you to swallow without deflecting it to the wind pipe?*

Mechanism of respiration in human beings

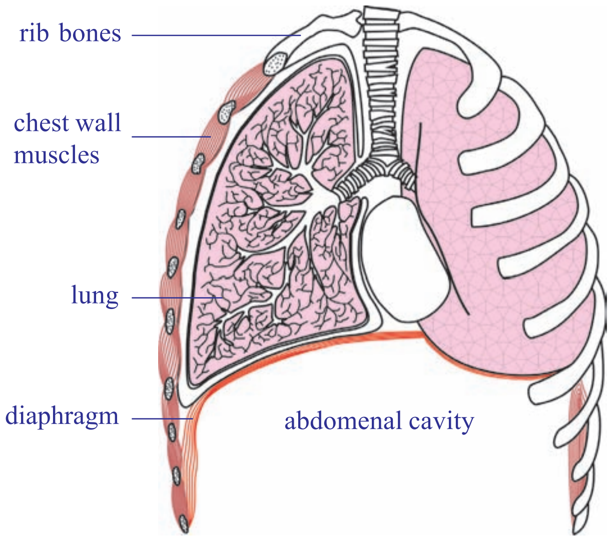


Fig-6: Diagram showing lungs and diaphragm

We know that breathing is the process of inhaling and exhaling. The organs involved are mainly the lungs. You can't see your lungs, but it's easy to feel them in action. Put your hands on your chest and breathe in very deeply. You will feel your chest getting slightly bigger. Now breathe out the air, and feel your chest return to its regular size. You've just felt the power of your lungs! The lungs themselves can neither draw in air nor push it out. Instead, the chest wall muscles and another flexible flattened muscle called diaphragm helps the lungs in moving air into and out of them. See how the diaphragm works in the Fig-7.

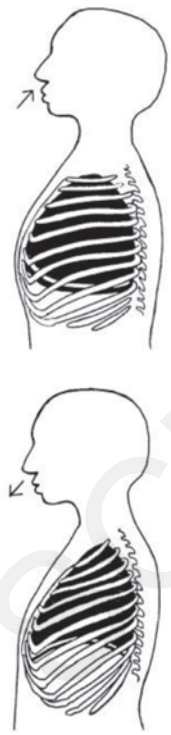


Fig-7: Movement of rib cage during inhalation, exhalation

- *What is the role of diaphragm and ribs in respiration? Are both active in men and women?*

The chest wall is made up of ribs, muscles, and the skin. The ribs are attached at an angle to the spine (if you run your finger along one of your ribs, you will notice that it extends downward from the spine). When we inhale, the chest wall moves up and expands. This causes an increase in the volume of the chest cavity.

The diaphragm may be imagined as the 'floor' if you think of the chest cavity as a "room." When the diaphragm is relaxed when we breath out, it is in the shape of a dome with the convex side of the dome extending into the chest cavity. When the diaphragm contracts during inhalation it flattens out a bit or the dome moves downward. As a result, the volume of the chest cavity is increased.

When the diaphragm flattens and the volume of the chest cavity is increased, its internal pressure decreases and the air from the outside rushes into the lungs. This is **inspiration** (inhalation).

Then the reverse occurs. The chest wall is lowered and moves inward, and the diaphragm relaxes and assumes its dome shape. These changes increase the pressure on the lungs; their elastic tissue contracts and squeezes the air out through the nose to the external atmosphere. This is **expiration** (exhalation).

Do you know?

Our lungs are spongy in nature. They are not of the same size. The lung towards left is slightly smaller making space for your heart! Lungs are protected by two membranes called pleura. A fluid filled between these membranes protects the lungs from injury and also aid in the expansion of the spongy and elastic lung muscle, as they slide one over the other.

You must have noticed that your own breathing is slow and shallow when you are at rest. It is deeper and faster when you exercise hard. Indeed, patterns of breathing show a great range, for they are coordinated with moment-by-moment needs of the body for supply of oxygen and removal of carbon dioxide.

What other situations affect your breathing?

It has been found that all movements of breathing stop at once when the nerves leading from the brain to the respiratory muscles are cut.

- *What can be concluded from this?*
- *What happens during the process of breathing?*
- *Which gas needs to be removed from our body during exhalation? Where does the extra amount of gas come from?*
- *What is the composition of inhaled air?*
- *When exhaled air is compared with inhaled air, is there any difference in composition?*

Gaseous Exchange (alveoli to capillaries)

Gaseous exchange takes place within the lungs by diffusion from the alveoli to blood capillaries and vice versa. The carbondioxide in the blood is exchanged for oxygen in the alveoli. These tiny air sacs in the lungs are numerous and only one cell thick. They are surrounded by capillaries that are also only one cell thick. Blood, dark red in colour flows from the heart through these capillaries and collects oxygen from the alveoli. At the same time, carbondioxide passes out of the capillaries and into the alveoli. When we breathe out, we get rid of this carbondioxide. The bright red, oxygen-rich blood is returned to the heart and pumped out to all parts of the body.

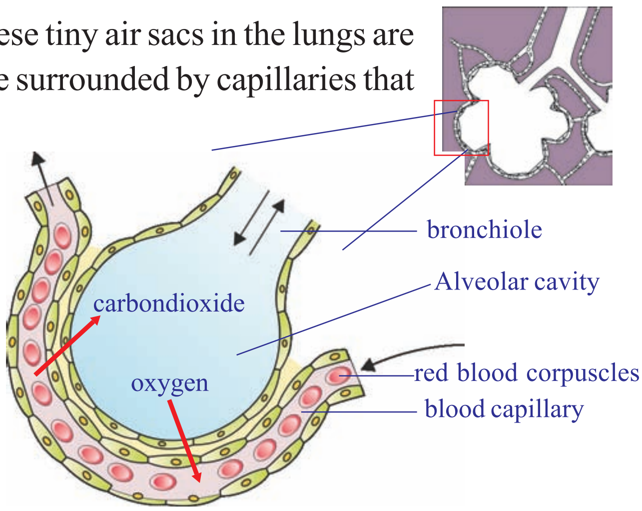


Fig-8: Alveolus with pulmonary capillary

As a result of gaseous exchange, the composition of inhaled and exhaled air is different. See the table given below. Approximate values are given in the table

Gas	% in inhaled air	% in exhaled air
Oxygen	21	16
Carbondioxide	0.03	4.4
Nitrogen	78	78

- *Why does the amount of Oxygen vary between exhaled and inhaled air?*
- *What has raised the percentage of carbon dioxide in exhaled air?*

Do you know?

The total lung capacity of human being is nearly 5800ml. Normally at rest who inhale or exhale approximately 500ml of air. 1200ml of air remains in lungs after complete exhalation.

Recall the activity of lung capacity performed by you in class VII in the chapter 'Respiration in Organisms'.

Transportation of gases

We know that air is a mixture of gases, that fills the lungs and the alveoli when that enters our body. The relative amount of different gases in air and their combining capacity with haemoglobin and other substances in blood determine their transport via blood in the body.

When oxygen present in the air is within normal limits (around 21%) then almost all of it is carried in the blood by binding to haemoglobin, a protein (quite like chlorophyll, the only major difference being it has iron in place of magnesium as in chlorophyll) present in the red blood cells. As oxygen is diffused in the blood, it rapidly combines with the haemoglobin to form oxyhaemoglobin. Not only haemoglobin can combine with oxygen, but the reverse can also happen to yield a molecule of haemoglobin and oxygen.

Carbon dioxide is usually transported as bicarbonate, while some amount of it combines with haemoglobin and rest is dissolved in blood plasma. Study the following equation for better understanding.

$$Hb + 4O_2 \longrightarrow Hb(O_2)_4 \text{ (in lungs)}$$

$$Hb(O_2)_4 \longrightarrow Hb + 4O_2 \text{ (in tissues)}$$

Do you know?

If haemoglobin is exposed to air at sea level, nearly every molecule combines with oxygen to form oxyhaemoglobin. At a height of 13 km (about 8 miles) above sea level, the concentration of oxygen is much lower about one fifth at sea level.

Fig-9: Mountaneer

Under these conditions only about half as many molecules of oxygen combine with haemoglobin to form oxyhaemoglobin. This is important, because blood cannot carry enough oxygen to the tissues if haemoglobin is combined with few oxygen molecules. In fact, human life is impossible at such an altitude without a supplementary supply of oxygen. Provision for such a supply is built into modern aircraft, which have pressurized cabins that maintain an enriched air supply. We would face other type of problems when we go deep into the sea.

Gaseous exchange (capillaries to cells and back)

In the capillaries over the tissues, haemoglobin meets a very different environment. The tissue cells are continuously using oxygen, hence, the concentration of oxygen is quite low in them. It might be only one third of that in the lungs. As the concentration of oxygen is so low, oxyhaemoglobin releases the oxygen molecule that enters the cells. In the reactions that occur within cells in our bodies, carbon dioxide and water are produced and energy is released to be used up for different purposes.

Cellular respiration

The term cellular respiration refers to the pathway by which cells release energy by breaking the chemical bonds of glucose molecules. Thus cellular respiration is the essential processes of life. So living cells must carry out cellular respiration. It can be in the presence of oxygen that is 'aerobic respiration', or in the absence of it, the respiration is called 'anaerobic respiration'. In Animals the anaerobic respiration leads to formation of lactic acid from glucose. In anaerobic repiration few ATP molecules are produced. Cellular respiration in prokaryotic cells like that of bacteria occurs within the cytoplasm. In eukaryotic cells cytoplasm and mitochondria are the sites of the reactions. The produced energy is stored in mitochondria in the form of ATP. That is why mitochondria are called "power houses of the cell".

How does this affect the energy release? As the change in the chemical nature of the molecule from one stage to the next is meagre, small amount of energy is released. The complete breakdown of a sugar molecule with the release of all its available energy involves a series of different chemical reactions.

From the breakdown of glucose, the energy is released and stored up in a special compound, known as ATP (adenosine triphosphate). It is a small amount of chemical energy.

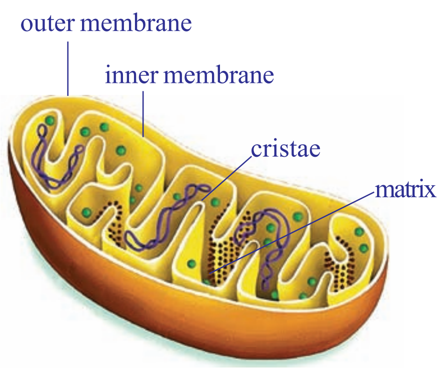


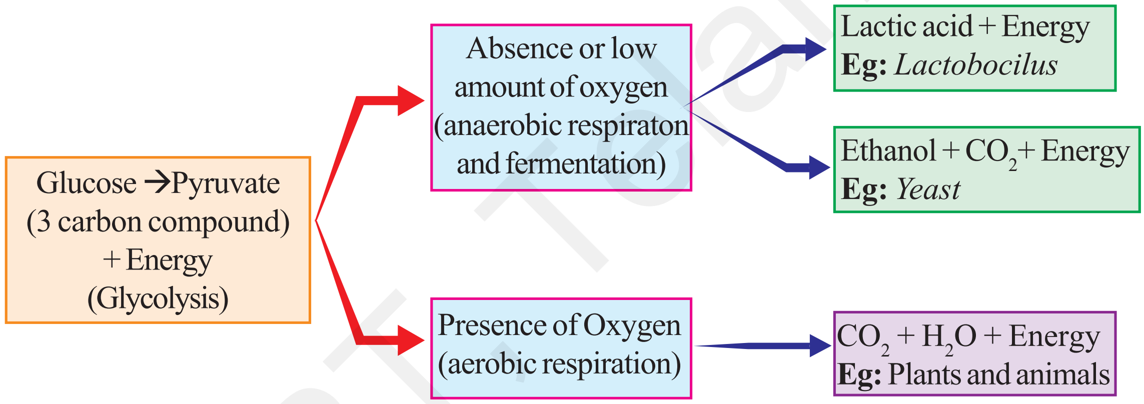
Fig-10: Mitochondria

The energy currency of these cells is ATP an energy rich compound that is capable of supplying energy wherever needed within the cell. Each ATP molecule gives 7200 calories of energy. This energy is stored in the form of phosphate bonds. If the bond is broken the stored energy is released.

- *Do cells of alveoli or lungs also require oxygen to carry out cellular respiration? Why/Why not?*

The exact chemical details of the breakdown of sugar or other foods within a living cell does not take place as a single reaction, but occurs in a series of small steps.

In short, at cellular level we could have the following pathways starting with glucose (It is one example, remember that there are other components of food as well).



Glucose is the most commonly used sugar for deriving energy in plants, animals and in microorganisms. In all these organisms the glucose is oxidized in two stages. In the first stage it is converted into two molecules of pyruvic acid. In the second stage if oxygen is available pyruvic acid is oxidized to CO_2, water and large amount of energy is released.

If oxygen is inadequate or not utilised, pyruvic acid is converted into ethanol. This process is called fermentation. In some bacteria lactic acid is formed and very little amount of energy (nearly one tenth of that is produced with adequate amount of oxygen) is released.

Can energy be released without oxygen?

- *After undergoing strenous exercise, we feel pain in muscles. Does adequate oxygen reach the muscles?*
- *What is being formed in the muscles?*

When you sprint a hundred yards, you do a considerable amount of muscular work. But you do not start a race by standing on the track and panting for a few minutes to stoke up with oxygen first. In fact you can run the race with very little extra breathing. The fastest sprinters do not breathe at all when running a hundred yards. After you have reached the destiny, however, you feel very different. Depending on your state of training, and on how hard you ran, you will pant for some minutes after the race, until your breathing gradually returns to normal.

Fig-11: Athlete (Strenous excercise)

These facts could be linked up with what we have learned so far about ATP. It might be that the race was run on the energy produced when the ATP already present in your muscles was being converted to ADP. Unfortunately this pleasantly simple idea is inadequate, because we only carry sufficient ATP in a muscle to last for about half a second when doing vigorous exercise. There must be some other explanation for the way in which we can produce energy first and then use up oxygen later.

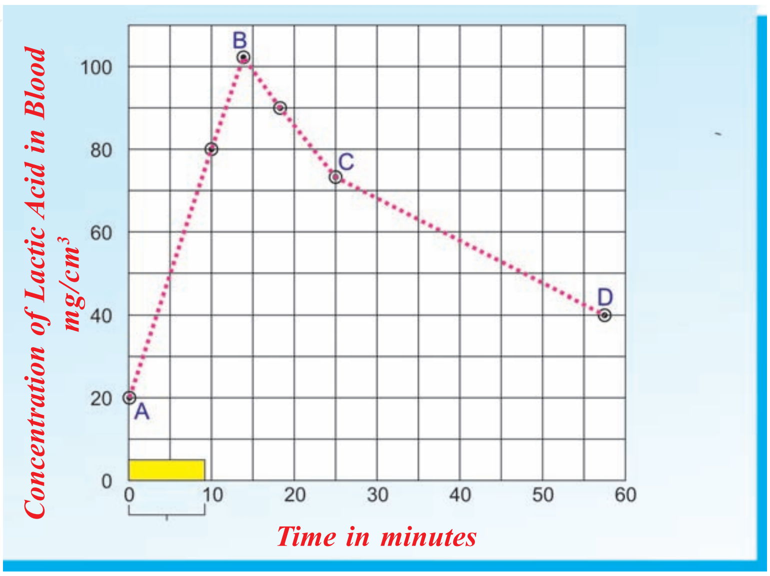
One approach in the study of this problem was to analyze the blood of a person during and after exercise. For obvious reasons the athlete taking part in the case study conducted had to stay still where the equipment was. He pedaled a stationary bicycle, or ran on a tread mill (belt moving as fast backwards as the athlete moved forwards). Some results are shown in the graph. Vigorous exercise lasted for nine minutes (shown by the bar at the base of the diagram) and regular blood samples were taken and analyzed. One particular compound in the blood, lactic acid, was found to vary greatly in its concentration as you can see from the graph.

Observe the graph of lactic acid accumulation in the muscles of an athelete and answer the following questions.

(Graph showing varying amount of lactic acid in the muscles)

x - axis = Time in minutes

y - axis = Concentration of lactic acid in blood mg/cm^3



Graph showing effects of vigorous excercise on the concentration of lactic acid in blood.

a. What was the concentration of lactic acid in the blood to start with?

b. What was the greatest concentration reached during the case study?

c. If the trend between points C and D were to continue at the same rate, how long might it take for the original lactic acid level to be reached again? (Hint: extend the line CD until it reaches the starting value.)

d. What does high level of lactic acid indicate about the condition of respiration?

Accumulation of lactic acid results in muscular pain. If we take walk, brisk walk, slow jogging, running for same distance we feel that there an increase in pain levels this is because of lactic acid accumulation.

It seems as if the lactic acid was being produced rapidly by the active muscles, and then only gradually removed from the blood after exercise. What is surprising is that the athlete needs a great length of time to recover. The simplest explanation we can produce at-this stage is that the glucose in the working muscles was being changed to lactic acid. The energy stored in lactic acid molecules is less than that in glucose molecules, and if the acid comes from the glucose then the energy released could be used to rebuild ATP from ADP and phosphate.

During a 100 m race a well-trained athlete can hold his breath all the time it is not until afterwards that he pants. In this case, the muscles are using the energy released during the anaerobic breakdown of glucose. It is not until afterwards that the athlete obtains the oxygen needed in order to remove the lactic acid. Therefore, when we under-take strenuous exercise we build up what is called an ***oxygen debt*** which has to be repaid later. In a longer race athletes have to breathe all the time, so some lactic acid is removed while they are running, and they can go on for longer before becoming exhausted. The presence of lactic acid in the blood is the main cause of muscle fatigue, but if the body is rested for long enough the tiredness goes.

Anaerobic respiration

We have found that living things produce carbon dioxide and give out energy. If these processes are caused by an oxidation process, what happens if the oxygen supply is cut off? If human muscles can go on releasing energy when they are short of oxygen, what can the cells of other living organisms do?

Let us find out by doing some experiments.

Lab Activity

Experiments with yeast

To test this idea we can see whether it is possible to detect any rise in temperature and the production of carbon dioxide, when living organisms are kept away from a supply of oxygen.

Yeast grows rapidly if it is supplied with glucose in solution. Indeed, wild yeasts are normally found growing on the skins of fruits like grapes and apples, from which they derive their food supplies. Our immediate problem is to remove the oxygen from the glucose solution and yeast.

Arrange the apparatus as shown in fig. 12

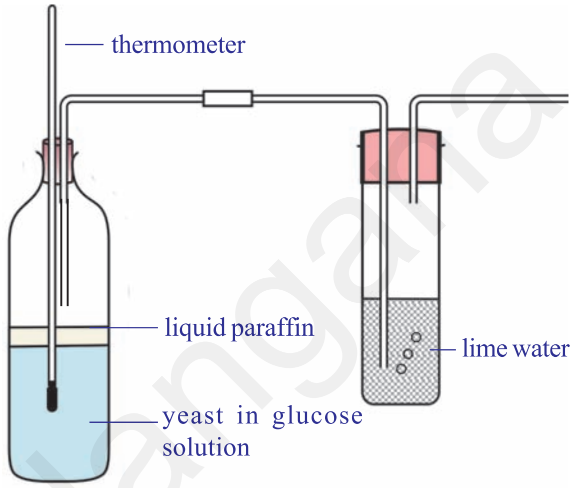


Fig-12: Testing for production of heat and CO_2 under anaerobic respiration

1. You can remove dissolved oxygen from glucose solution by heating it for a minute, and then cooling it without shaking. Now put in some yeast; the supply of oxygen from the air can be cut off by pouring one centimetre layer of liquid paraffin on to the mixture.
2. If you wish to check that the oxygen has been removed from the mixture, add a few drops of diazine green (Janus Green B) solution to the yeast suspension before you pour the liquid paraffin (wax) over it. This blue dye turns pink when oxygen is in short supply around it.
3. Another bottle containing bicarbonate/indicator solution (or lime water) is attached as in the Fig-12.

During the process of anaerobic respiration some amount of carbondioxide is released. The released carbondioxide passes through the tube and turns lime water milky. We can observe the change in the temperature also during this process out for yourself. You may prefer to carry out the 'carbon dioxide production' part of the experiment on a smaller scale, using test tubes. If you do, then warm them to about 37^{o} C in order to speed up the test.

- *What happens when a baker prepares a dough by mixing yeast in it?*

Fermentation

Let us recall maida dough and yeast powder activity that you performed in class VIII in the chapter 'The story of microorganisms'. Why volume of the dough has increased? Which gas released in that reaction?

If yeast and sugar solution are left to stand without oxygen for some days, they develop a characteristic smell, caused by production of new compound called ethanol, which has been manufactured by the yeast from the sugar. The same type of smell you can notice from preserved idly, dosa dough at your home, but not in refrigirator. The ethanol can be separated from the yeast-glucose mixture by the process of fractional distillation since ethanol boils at a lower temperature (70°C) than the sugar solution. Collect information about fractional distillation with the help your teacher.

Quite like aerobic respiration this is a process of producing energy when there is no supply of oxygen.

- *Respiration is an energy releasing pathway, do you agree? Justify your answer.*

Respiration versus combustion

Lavoisier around the late 18thcentury, through a series of carefully performed experiments, came to the opinion that respiration was a process like combustion. He wrote in a compilation in 1783, "respiration is a combustion process. It is a very slow process and here oxygen is not only combines with carbon but also with hydrogen." Robinson also stated that respiration is a type of combustion and combustion is the source of heat in animals.

Activity-2

Observing changes during combustion of glucose

Arrange apparatus as shown in the Fig-13 and heat the test tube over a flame. Does the glucose in the test tube melt? What happens if you heat for some more time?

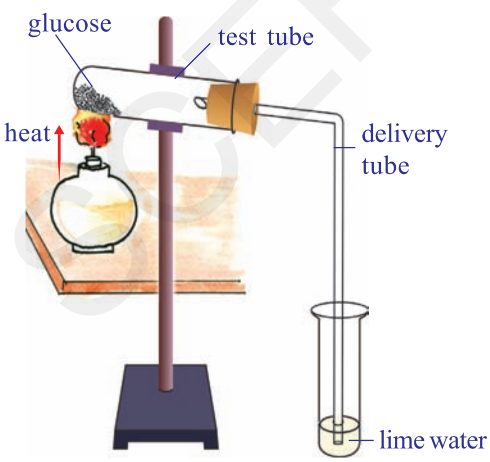


Fig-13: CO_2 - by-product of energy release

When glucose burns, carbon dioxide and water are produced and energy is released as heat.

We know that combustion of glucose gives us carbon dioxide, water and energy while from the respiratory equation we get the same products. But essentially the processes should differ due to following reasons.

1. Glucose must be burnt at high temperature in the laboratory to liberate energy, if it happened in our cells,all cells would be burnt.
2. Once glucose starts burning we can't stop the process easily, but living cells are able to exercise control over the sort of burning of glucose in the presence of oxygen.

3. Water normally stops combustion from taking place while cells contain a lot of water and respiration still goes on.

   What can you conclude from this?

Heat production by living organisms

Heat production was a feature of burning glucose or sugar as you observed earlier. Living animals and plants usually produce energy in the form of heat.

We feel warm when we wear sweater in winter season. We know that sweater prevents loss of heat energy produced by the body. Does this suggest any way in which our bodies lose heat to the surroundings?

- *What are the other ways in which our body loses heat?*

Heat is constantly lost from the body surface thus it must be continuously generated within our bodies to replace what has been lost to keep the body temperature constant.

- *Is the rate of heat production always the same?*

In the course of vigorous activity, a greater amount of heat is generated. We know that we feel hot after some form of strenuous exercise such as running.

During cellular respiration energy is released. Some amount of energy is stored in the form of ATP. Some amount of energy is utilised in our day to day life activities. And the excess amount of energy is released in the form of heat. But in case of vigorous activity like running we need more amount of energy. For this the rate of respiration is increased. So heat is also released in excess quantity. That's why we feel warm. If the oxygen is not sufficient during vigorous exersise muscles start anaerobic respiration. Hence, lactic acid is formed. We know that accumulation of lactic acid causes pain in muscle. We reach normal position after some rest. Deep breathing helps us to restore energy in our body. Refer in annexure about Pranayama.

Evolution in gases exchanging system

Exchange of gases is a common life process in all living organisms, but it is not same in all. Single celled organisms Amoeba or multicellular organisms like *Hydra, Planaria,* roundworms and earthworms obtain oxygen and expel carbon dioxide directly from the body by the process of diffusion. In other multicellular animals special organs are evolved. Animals either terrestrial or aquatic adapted to different types of respiration and possess different types of respiratory organs mostly depending on the habitat in which they live. Body size, availability of water and the type of their circulatory system are some of the reasons for the animals to develop different types of respiratory organs.

There is tracheal respiratory system in insects like cockroach, grasshopper etc. Tracheal respiratory system consists of series of tubes called trachea. This is divided into fine branches called tracheoles which carry air directly to the cells in the tissues.

Some aquatic animals like fishes have developed special organs for respiration which are known as gills or branchiae. Blood supplied to gills through capillaries which have thin walls where gases are exchanged. This is called branchial respiration. Fish keeps its mouth open and lowers the floor of the oral cavity. As a result water from outside will be drawn into the oral cavity. Now the mouth is closed and the floor of the oral cavity is raised. Water is pushed into the pharynx and is forced to gill pouches through internal branchial apertures. Gill lamellae are bathed with water and gaseous exchange takes place.

Respiration through skin is called cutaneous respiration. Frog an amphibian can respire through cutaneous and pulmonary respiration (through lungs) processes as well as Bucco-Pharyngeal Cavity. Terrestrial animals like reptiles, birds and mammals, respire through lungs. Ask your teacher how crocodiles and dolphins respire?

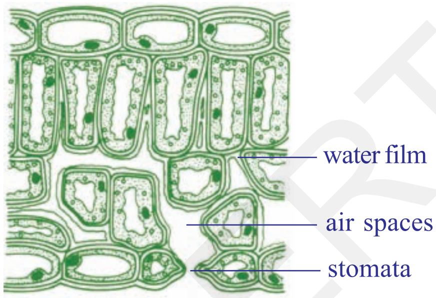


Fig-14: Leaf as a respiratory organ

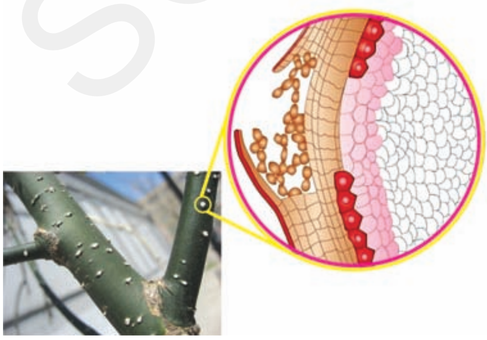


Fig-15: Lenticels on stem

Respiration in Plants

You already know about stomata in leaf where gaseous exchange takes place in plants. There are other areas on the plant body as well through which gaseous exchange take place like surface of roots, lenticels on stem etc. (Fig showing stomata and lenticels). Some plants have specialized structures like breathing roots of mangrove plants as well as the tissue in orchids that produces oxygen is also required by plants to produce energy and carbon dioxide is released. But CO_2 is required elsewhere in the plants. Try to identify them.

Conduction within the plant

The stomatal openings lead to a series of spaces between the cells inside the plant. Which form a continuous network all over the plant. The spaces are very large in the leaves, much smaller in other parts of the plant. The air spaces are lined with water where the oxygen is dissolved in this and passes through the porous cell walls into the cytoplasm. Here the glucose is broken down into carbon dioxide and water with the liberation of the energy. The carbon dioxide passes out into the air spaces by a similar method.

The whole system works by diffusion; as the oxygen is used up by the cells a gradient develops between the cells and the air in the spaces. Similarly between the air in the spaces and the air outside the stomata and lenticels, so oxygen passes in. In the same way, as more carbon dioxide is released by the cells a gradient occurs in the reverse direction and it passes out to the environment.

Aeration of roots

Fig-16: Aerial/respiratory roots

Most plants can aerate their roots by taking in the oxygen through the lenticels or through the surface of their root hairs (as their walls are very thin). They obtain oxygen from the air spaces existing between the soil particles. But, plants which grow in very wet places, such as ponds or marshy soils, are unable to obtain oxygen. They are adapted to these water-logged conditions by having much larger air spaces which connect the stems with the aerial roots, making diffusion from the upper parts much more efficient.

The most usual adaptation is to have a hollow stem. Next time when you are by a pond or marsh cut the stems of some of the plants which are growing there and see how many are hollow compared with a similar number of species of plants growing in normal soil. The problem of air transport is more difficult for trees and not many survive with their roots permanently in water. An exception is the mangrove tree of the tropics which have aerial roots above the soil surface and takes in oxygen through these roots.

To know more about respiration in plants we should perform the following activities.

Activity-3

Take a handful of moong or Bengal gram seeds. Soak the seeds in water a day before. Keep these soaked seeds in a cloth pouch and tie with a string tightly. Keep the cloth pouch in a corner of your class room. Next day collect the sprouts/ germinated seeds from the pouch, keep it in a glass bottle/plastic bottle(around 200 ml capacity). Take a small beaker, fill three fourth

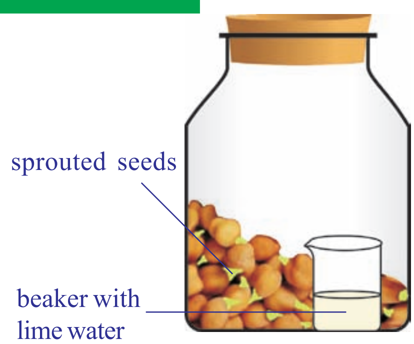


Fig-17: Evolved CO_2 in respiration

with lime water. Keep it in the beaker carefully. Close the plastic bottle tightly. Make a similar set with unsprouted seeds. Keep this set undisturbed for one or two days. During this time observe the colour of lime water in both the sets. In which set does the colour change faster? Why?

Activity-4

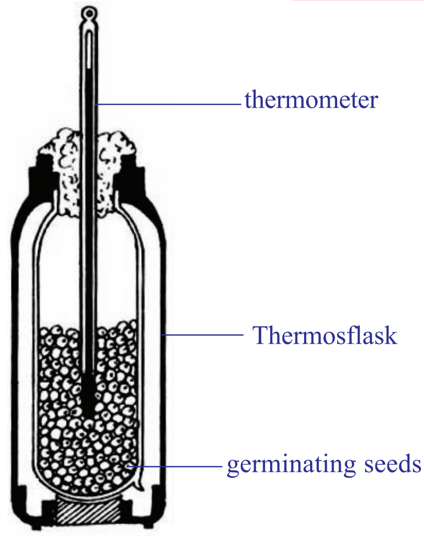


Fig-18: Heat evolved during respiration

Take sprouts which were prepared for above activity in a thermosflask. Remove the lid and prepare a cork (with thermocol, or rubber or any other material) through which you can bore a hole to insert a thermometer. Take care that the bulb of the thermometer should dip in the sprouts. Close the flask with this tight fitting cork. Record the temperature for every two hours. You are advised to do this for at least 24 hours.

- Make a graph by using your observations.
- *Is there any increase in temperature?*
- *Does the temperature increase steadily or does it abruptly increase at a time of the day?*
- *Where does the heat come from?*

Photosynthesis versus Respiration

Plants carry out photosynthesis, which means that they produce their own food from atmospheric CO_2 using light energy from the sun. This process is a complex series of steps involving the conversion of light energy into chemical energy, which is then used to synthesize sugars from carbon dioxide. This is a process of synthesis or an anabolic process which occurs in the chloroplasts.

The equation below summarizes the photosynthetic process

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light energy}} \underset{\text{glucose}}{C_6H_{12}O_6} + 6H_2O + 6O_2$$

Once produced, the sugars can then be used for the process of respiration to provide energy to run all life processes.

Respiration as we know is not just the exchange of gases. It is the process of breakdown of complex food molecules or a catabolic process to produce chemical or potential energy.

This can be summarized by the equation

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + \text{Energy}$$

Photosynthesis and respiration appear to be opposing reactions, but both have very different biochemical pathways and are essential for a plant's metabolism.

Photosynthesis takes place in the chloroplast to produce sugars, starch and other carbohydrates for the plant's metabolic needs. Cellular respiration occurs in mitochondria where these carbohydrates are "burned" to produce chemical energy to function at the cellular level. During day time, the rate of photosynthesis is usually higher than that of respiration while at night it is just reverse in most plants. Temperature, humidity, light intensity etc. seem to affect the ratio of photosynthesis and respiration in plants.

Key words

Aerobic respiration, Anaerobic respiration, Alveoli, Pharynx, Trachea, Bronchi, Bronchioles, Epiglottis, Anabolic process, Catabolic process, Aerial roots, Lenticels, Fermentation, Energy currency

What we have learnt

- By "respiratory system" we usually mean the passages that transport air to the lungs and to the microscopic air sacs in them, called alveoli (where gases are exchanged) and vice versa.
- The term 'respiration' refers to the whole chain of processes from the inhalation of air to the use of oxygen in the cells.
- Lavoisier found that the air that we breathe out precipitated lime water
- Air passes from nostrils to nasal cavity to pharynx, larynx, trachea, and bronchi, bronchioles to alveoli and blood and back through the same route.
- Gas exchange in the lungs takes place in the tiny air sacs called alveoli in the lungs. The lungs have millions of alveoli and each lies in contact with capillaries.
- Diaphragm is a muscular tissue present at the floor of the chest cavity.
- During inspiration (inhalation) the volume of the chest cavity is increased as the diaphragm contracts and dome flattens out, its internal pressure decreases and the air from the outside rushes into the lungs.
- During expiration, the chest wall is lowered and moves inward, and the diaphragm relaxes and assumes its dome shape. These changes increase the pressure on the lungs; their elastic tissue contracts and squeezes the air out through the nose to the external atmosphere.
- Air is filtered in the nasal cavity and the whole length of the trachea.
- The moist surface of the lining of the nasal cavity, and the hairs growing from its sides, remove some of the tiny particles of dirt in the air. In addition, as the inhaled air passes through the nasal cavity, its temperature is brought close to that of the body, and it takes up water vapour. So that it becomes more moist than before.

- Pharynx is a common passage of digestive and respiratory system. Epiglottis, a flap like muscular valve controls movement of air and food towards their respective passages.
- Larynx is a stiff box like structure containing our vocal cords. When air passes out of the lungs and over the vocal cords, it causes them to vibrate. This produces sounds on the basis of our speech, song etc.
- Trachea is the wind pipe channeling air to lungs.
- At its lower end the trachea or the wind pipe divides into two bronchi-one leading to each lung.
- The bronchi divide into smaller and smaller branches called bronchioles.
- These finally terminate in clusters of air sacs called alveolus in the lungs which are very small and numerous. Gaseous exchange takes place here as blood capillaries take up oxygen and expel carbon dioxide here.
- Aerobic respiration occurs in adequate supply of air producing a lot of energy, carbon dioxide and water.
- Anaerobic respiration and fermentation occurs in inadequate supply or absence of oxygen to produce energy.
- Cells may resort to the breakdown of 3 carbon compound, pyruvate, aerobically or anaerobically depending upon the availability of oxygen. Usually in multicellular organisms cells fail to carry on the process of anaerobic respiration for long.
- Respiration is not essentially a process of combustion differ due to following reasons
  - Glucose must be burnt at high temperature in the laboratory to liberate energy, if it happened in our cells, all cells would be burnt.
  - Once glucose starts burning we can't stop the process easily, but living cells are able to exercise control over the sort of burning of glucose in the presence of oxygen.
  - Water normally stops combustion from taking place while cells contain a lot of water and respiration still goes on.
- Photosynthesis and respiration appear to be opposing reactions, but both have very different biochemical pathways and are essential for a plant's metabolism.
- Photosynthesis takes place in the chloroplast to produce sugars, starches and other carbohydrates for the plant's metabolic needs.
- Cellular respiration occurs in mitochondria where mainly these carbohydrates are "burned" to produce chemical energy to do work at the cellular level.

Improve your learning

D5X2Q7

1. Distinguish between (AS1)
   a) inspiration and expiration
   b) aerobic and anaerobic respiration
   c) respiration and combustion d) photosynthesis and respiration
2. State two similarities between aerobic and anaerobic respiration.(AS1)
3. Food sometimes enters the wind pipe and causes choking. How does it happen?(AS1)
4. Why does the rate of breathing increase while walking uphill at a normal pace in the mountains? Give two reasons.(AS1)
5. Air leaves the tiny sacs in the lungs to pass into capillaries. What modification is needed in the statement?(AS1)
6. Plants photosynthesize during daytime and respire during the night. Do you agree to this statement? Why? Why not?(AS1)
7. Why does a deep sea diver and mountainair carry oxygen cylinder on her back?(AS1)
8. How are alveoli designed to maximize the exchange of gases?(AS1)
9. Where will the release of energy from glucose in respiration take place? Mala writes lungs while Raziya writes muscles. Who is correct and why?(AS1)
10. What is the role of epiglottis and diaphragm in respiration?(AS1)
11. How does gaseous exchange takes place at blood level (or) cellular level?(AS1)
12. Explain the mechanism of gaseous exchange at branchiole level.(AS1)
13. After a vigorous excercise or work we feel pain in muscles. What is the relationship between pain and respiration?(AS1)
14. Raju said stem also respire along with leaves in plants. Can you support this statement? Give your reasons.(AS1)
15. What happen if diaphragm is not there in the body?(AS2)
16. If you have a chance to meet pulmonologist what questions are you going to ask about pulmonory respiration?(AS2)
17. What procedure do you follow to understand anaerobic respiration in your school laboratory?(AS3)
18. What are your observations in combustion of sugar activity?(AS3)
19. Collect information about cutaneous respiration in frog. Prepare a note and display them in your classroom.(AS4)
20. Collect information about respiratory diseases (because of pollution, tobacco) and discuss with your classmates.(AS4)
21. What is the pathway taken by air in the respiratory system? Illustrate with a labelled diagram.(AS5)
22. Draw a block diagram showing events in respiration. Write what you understood about cellular respiration.(AS5)

23. How you appreciate the mechanism of respiration in our body?(AS6)
24. Prepare an article on anaerobic respiration to present school symposium.(AS7)
25. Prepare a cartoon on discussion between haemoglobin and chlorophyll about respiration.(AS7)

Fill in the blanks

1. Exhaled air contains __________ and __________.
2. A flap like muscular valve controls movement of air and food is ___________.
3. Energy currency of the cell is called _____________.
4. Lenticels are the respiratory organs exists in __________ part of plant.
5. Mangroove trees respire with their ____________.

Choose the correct answer

1. We will find vocal cords in ( )

   A) larynx B) pharynx C) nasal cavity D) trachea

2. Cluster of air sacs in lungs are called ( )

   A) alveoli B) bronchi C) bronchioles D) air spaces

3. Which of the following is correct ( )

   i) the diaphragm contracts - volume of chest cavity increased

   ii) the diaphragm contracts - volume of chest cavity decreased

   iii) the diaphragm relax - volume of chest cavity increased

   iv) the diaphragm relax - volume of chest cavity decreased

   A) i B) i & ii C) ii & iii D) iv

4. Respiration is a catabolic process because of ( )

   A) breakdown of complex food molecules B) conversion of light energy

   C) synthesis of chemical energy D) energy storage

5. Energy is stored in ( )

   A) nucleus B) mitochondria C) ribosomes D) cell wall

Annexure

Pranayama - The art of breathing

It is wonder to know that only human beings have to learn how to breath. Our lungs are divided into lobes. At each breath we will inhale or exhale only 500ml of air. Where as our lung capacity is approximately 5800ml. So most of the time breathing takes place in the upper lobes only. This means we are not using our lungs to their fullest capacity. Even after complete expiration approximately 1200ml of air remains in our lungs. So we can make use of 4600ml of lung capacity for breathing. Think what changes are required to inhale that much air.

The Indian ayurvedic physician. Patanjali developed a scientific breathing practice called *Yogabhyasa*. Maharshi Patanjali proposed a theory called Astanga yoga. He was introduced 195 yogic principles in eight divisions. 1. Yama (Social discipline), 2. Niyama (Individual discipline), 3. Asana (Body posture), 4. Pranayama (Expansion of vital energy), 5. Prathyahara (With drawal of senses), 6. Dharana (Concentration), 7. Dhyana (Meditation), 8. Samadi (Self realisation).

The art of breathing in Yogabyasa is called *Pranayama* prana means gas, ayama means journey. In Pranayama practice air is allowed to enter three lobes of lungs inorder to increase the amount of oxygen to diffuse into blood. Deep breaths in Pranayama helps us to reduce breathings per minute form 20-22 to 15. Because of these deep breaths more amount of oxygen available to brain and tissues of the body will be more active. It is very important to practice Pranayama regularly to make our life healthy and active. All people irrespective of age and sex should practice Pranayama under the guidance of well trained Yoga Teacher to improve the working capacity of lungs.

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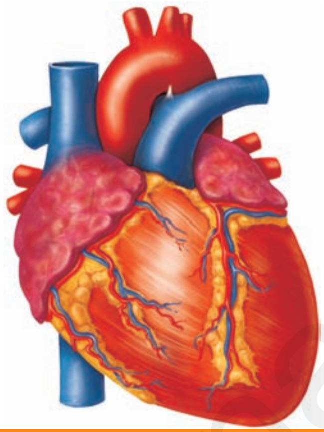


Chapter
3
Transportation

All living organisms need nutrients, gases, liquids etc., for growth and maintenance of their body.

All organisms would need to transport these materials to all parts of their body whether they are unicellular or multicellular.

In unicellular organisms these may not have to be transported to longer distances while in multicellular forms substances have to be transported to long distances as far as say around 100 feet for the tallest plant on earth!

In organisms like amoeba, hydra etc., all the materials are transported through simple processes like diffusion, osmosis etc.,

It would have taken long periods of time for movement of materials if large multicellular organisms (bodies made of trillions of cells) adopted processes like diffusion and osmosis only.

To avoid delay a separate system is needed to carry the materials much faster and more efficiently. This specialized system is called 'the circulatory system'.

We eat solids, drink liquids, and breathe gases.

- *Do you think that it is possible to transport all the three types of materials, through a single system?*

Let us study how these are transported in our body.

Have you ever observed a doctor holding the wrist of a patient and looking at his watch for a minute? What is it that he is trying to find out?

The doctor is trying to count the heart beat of the patient. Try counting your own heartbeat.

Activity-1

Keep your index and middle fingers on your wrist below the thumb as shown in the Fig-1.

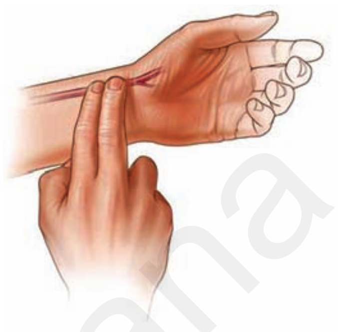


Fig-1: Pulse

You feel something pushing your fingers rhythmically up and down. Now let us count the rhythm which is called the pulse, for a minute. Now stand up and jog for one minute at the same place. Note the pulse for a minute. Take readings of at least two of your friends in the same manner and record in the following table.

Table-1

S.No	Name of the person	Pulse rate per minute	
		at rest	after jogging

- *What did you observe? Is the pulse rate same in both conditions?*

Activity-2

We see that pulse rate varies from person to person and situation to situation. When you are afraid or excited the pulse rate goes up. Observe your pulse rate while climbing stairs, running, etc. There is a relationship between the pulse rate and the beat of our heart. Now let us try to find out more about this relationship.

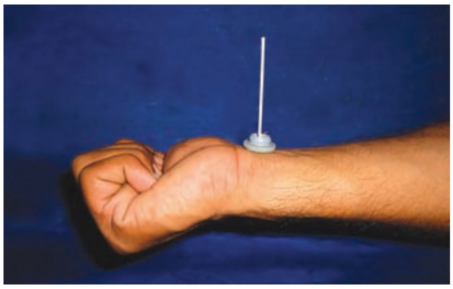


Fig-2: Pulse indicator

For this you have to make your own pulse indicator. Take an injection bottle lid/shirt button. Insert a matchstick as shown in Fig-2. Place it on your wrist. Observe movements in matchstick. Put your other hand on your chest and feel the beat of your heart.

- *Does the pulse rate indicator move with the beat of your heart?*

Do you know?

Variation in pulse rate

Newborn (0–3 months)	Infants (3-6 months)	Infants (6–12 months)	Children (1-10 years)	Children over 10 years & adults, including senior citizens	Well- trained adult athletes
100-150	90-120	80-120	70-130	60-100	40-60

In the year 1816, Rene Laennec invented the Stethoscope. Before the discovery of stethoscope doctors used to hear heart beat by keeping ear on the chest of the patient. Laennec found that paper tube helps to hear the heart beat perfectly. Then he used a bamboo instead of paper tube to hear heart beat. Laennec called it stethoscope.

Activity-3

Let us make our own Stethoscope.

Make a paper tube 10 inch long and one inch in diameter. Keep one end of it at your ear and the other end on the chest of your friend, so that your friend's heartbeat is audible to you. Listen carefully and count the heart beats for a minute.

Also count down your friend's pulse rate. Note observations of at least 10 students of your class in the following tabular form.

Table-2

S.No	Name of the student	Heart beat at rest/min	Pulse rate at rest
1	Eshwar	72	72

- *What is the relationship between the heart beat and the pulse rate?*

Now let us try to understand the structure and method of working of this vital organ, the heart. It is the beat of the heart which sends blood rushing to different parts of our body. Heart is located in between the lungs and protected by rib cage. The size of your heart is approximately the size of your fist.

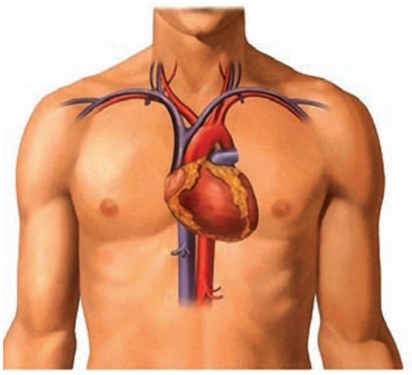


Fig-3: Location of Heart

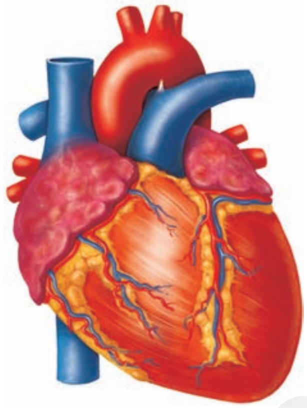


Fig-4: Heart External features

Lab Activity

Aim: Observation of the internal structure of the mammalian heart.

Material required: Freshly collected specimen of heart of sheep or goat from the butcher. Soda straws, used pen refils, sharp and long blade or scalpel, tray, a jug of water, Dissection scissors, forceps.

Since the structure of all the mammalian hearts are similar, we take the sheep's or goat's heart for our observation.

Procedure for observation:

- Before conducting the activity wash the heart thoroughly so that, blood is completely drained from the chambers of heart.
- Take soda straws and insert them into the stumps of the blood vessels. Note your observations as you proceed.
- *What is the shape of the heart?*
- *How many layers are covering the heart?*
  (Now remove the layers covering the heart, and observe)
- *How many large blood vessel stumps are attached to the heart?*
- *Which end of the heart is broader and which end is narrow?*

Observe the arrangement of blood vessels (coronary vessels) on the wall of the heart.

(In case you don't have a model or a goat's heart, look at the figures given carefully for observation)

Internal structure of the heart

- Keep the heart in the tray in such a way that a large arch like tube faces upwards. This is the ventral side.

* Now take a sharp blade or scalpel and open the heart in such a way that the chambers are exposed. Take the help of the Fig-5.

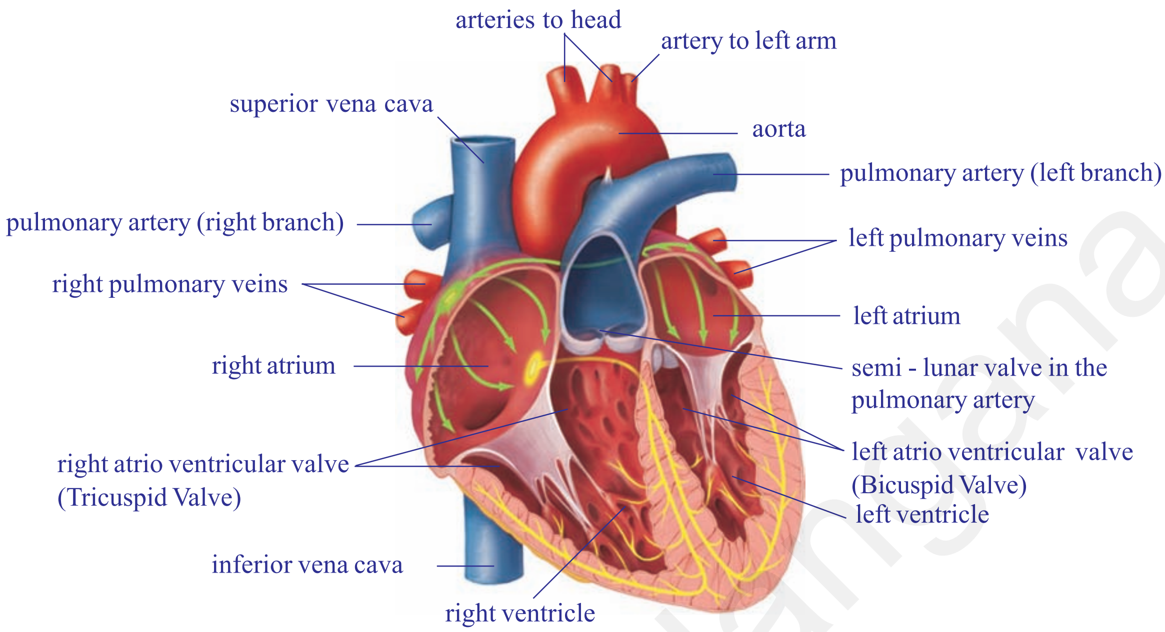


Fig-5: Internal structure of heart

Now observe the internal structure. Observe the wall of the heart.

* *Is the thickness of the wall of the heart uniform throughout?*
* *How many chambers are there in the heart?*
* *Are all the chambers of the same size?*
* *What other differences could you observe between the chambers?*
* *Do you find any specific observation in between two chambers?*
* *Are all the chambers connected to each other?*
* *How are they connected to each other? How are they separated?*

Write a note on your observations of the heart. Compare your notes with the description given below.

The heart is a pear shaped structure, triangle in outline, wider at the anterior end and narrower at the posterior end.

The heart is covered by two layered membrane. This is called pericardium (pericardial membrane). The space between these two layers is filled with pericardial fluid, which protects the heart from shocks.

The heart is divided into four chambers.

Two upper chambers are called atria (auricles), and the lower ones are called ventricles.

The blood vessels found in the walls of the heart are coronary vessels which supply blood to the muscles of the heart.

The walls of the ventricles are relatively thicker than the atrial walls.

Observe the presence of blood vessels attached to the heart.

- *How many blood vessels are attached to the heart?*
- *Are all the blood vessels rigid? How many of them are rigid?*
- *Does the rigidity of blood vessel have anything to do with circulation?*

Arteries are the blood vessels which originate from the heart and supply blood to various organs in the body. They have thick walls. The largest artery is aorta. The relatively small artery is pulmonary artery which carries blood from the heart to the lungs.

The veins are the blood vessels which bring blood from all body parts to the heart. Veins have relatively thin walls. The vein which is at the anterior end of the right side of the heart is superior venacava, which collects blood from anterior parts (head and neck) of the body.

The vein which brings blood from posterior parts of the the body (hands, legs etc.) is inferior venacava.

The two atria and the two ventricles are separated from each other by muscular partitions called septa. The openings between atria and ventricles are guarded by valves.

The valve present on the Right Auriculo-ventricular septum between Right atrium and Right Ventricle is referred to as "Tricuspid Valve". The valve present on the left Auriculo-ventricular septum between left atrium and left ventricle is referred as Bicuspid valve or Mitral valve. A major blood vessel that originates from right ventricle is Pulmonary aorta. The valves present at the region of Pulmonary aorta are called "Pulmonary valves". A major blood vessel that originates from left ventricle is systemic aorta. The valves present at the region of systemic aorta are called Systemic valves.

In the right atrium we can observe the openings of superior and inferior venacava. In the left atrium, we can observe the openings of pulmonary veins, that bring blood from lungs.

Aorta arises from the upper part of the left ventricle. It supplies oxygenated blood to the body parts. From the upper part of the right ventricle pulmonary artery arises that supplies de-oxygenated blood to the lungs. After careful examination we can observe valves in the pulmonary artery and aorta as well.

Blood vessels and circulation

Let us study how we came to know about the structure and functions of the blood vessels.

It was not until 16^{th} century that we really came to know how our blood vessels functioned. In 1574, an Italian doctor, Girolamo Fabrici, was studying the veins in the leg. He noticed that they had small valves in them that permited blood to flow towards heart. These were one-way valves that checked backflow of blood.

Movement of leg muscle helps in movement of blood through veins.

Fig-6: William Harvey

Everyone thought that the blood leaving the left ventricle always moved away from the heart for which Fabrici paid no attention. He missed the importance of his own discovery. William Harvey (1578-1657), an English doctor, went to Italy for further education under Fabrici.

Harvey dissected the hearts of dead people and studied the valves between each atrium and its ventricle. He noticed that they were one-way valves. They allowed the blood to flow from the atrium to the ventricle without any hinderance.

When the heart contracted, however, no blood in the ventricle could flow back into the atrium. Instead, all the blood was pushed out into the arteries.

Harvey began thinking about the valves his teacher, Fabrici, had discovered in the leg veins. They were one-way, and they forced the blood to move toward the heart.

He checked that by tying off and blocking different veins in animals. He found that the veins always bulged on the side of the block away from the heart. As though the blood was trying to flow toward the heart. This was true of all veins.

In the arteries, the blood bulged on the heart side of any block he put in, as though it were trying to flow away from the heart and couldn't move in the other direction.

Harvey now saw what was happening. The heart pushed blood in to the arteries and the blood returned by way of the veins. It did this for both ventricles. The blood had a double circulation. Starting from the right ventricle blood left by way of the arteries to the lungs and returned by way of veins to the left atrium, this is known as 'pulmonary circulation'. From there it flowed into the left ventricle and left by way of aorta to the rest of the body and returned by way of the veins to the right atrium. From here it moved into the right ventricle, this is known as 'systemic circulation'. As blood reaches the heart two time through pulmonary cycle and again through systemic, it is called 'double circulation'.

Harvey also showed that it was impossible to suppose that the blood was used up in the body and that new blood was formed. He measured how much blood the heart pumped in one contraction and also counted the number of contractions.

He found that in one hour, the heart pumped out a quantity of blood that was three times the weight of a man. The body couldn't use up blood and form new blood at such a rate. The same blood had to circulate and be used over and over again.

Harvey still had some problem. The smallest arteries and veins that could be seen had to be connected by vessels too small to see. Were they really there?

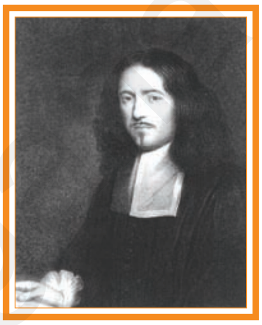


Fig-7: Marcello Malpighi

In the 1650s, scientists had learned to put lenses together in such a way that objects too small to see with the naked eye could be magnified and made visible. Marcello Malpighi (1628-1694), used the microscope to see tiny blood vessels that were invisible with naked eye.

Later, Malpighi studied the wings of bats. He could see blood vessels in their thin membranes under the microscope; he could see that the smallest arteries and veins were connected by very fine blood vessels.

He called these blood vessels "capillaries" from the Latin word for "hair", because they were as thin as the finest of hairs.

With the discovery of capillaries, the idea of the circulation of the blood was complete, and it has been accepted ever since.

Fig-8(a): Try like this

Now, we know that blood circulates in the blood vessels. But how did the scientists find out that blood moves in blood vessels? Is it possible to demonstrate the movement of blood in vessels without damaging the vessels?

Let us repeat the classical experiment to demonstrate the movement of blood in veins conducted by William Harvey in early 17th century, when there was no compound microscope or any other modern equipment.

1. Tie a tornquit just above the elbow of your friend as shown in Fig-8(b) whose blood vessels are prominent in the hand.
2. Ask him/her to hold a firm support as shown in the figure. Now the blood vessels can be seen more prominently.
3. Locate a prominent bluish blood vessel.
4. At the end of the vessel farthest from the elbow apply steady pressure, so as to close its cavity.
5. Now apply pressure from elbow towards the palm slowly and observe the change in the blood vessel. (Take the help of the figures given here.) Observe changes and discuss in your class.

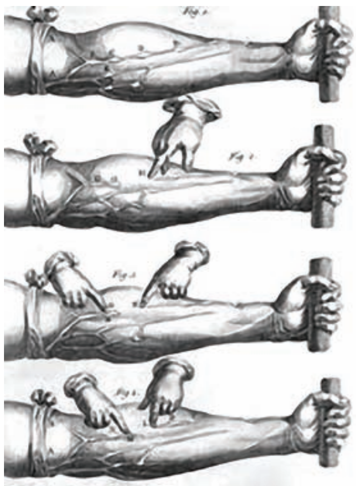


Fig-8(b): Harvey's demonstration

6. Remove the pressure.
7. Apply pressure from palm towards elbow and see what happens.

Answer the following:

- *Why do blood vessels (just below our skin) bulge on the side away from the heart when the hand is tied? What do you understand from it?*

Arteries and Veins

Arteries carry blood from the heart to body parts. Veins carry blood from body organs to heart. Let us observe the structural and functional differences between arteries and veins [see Fig-9(a) and 9(b)].

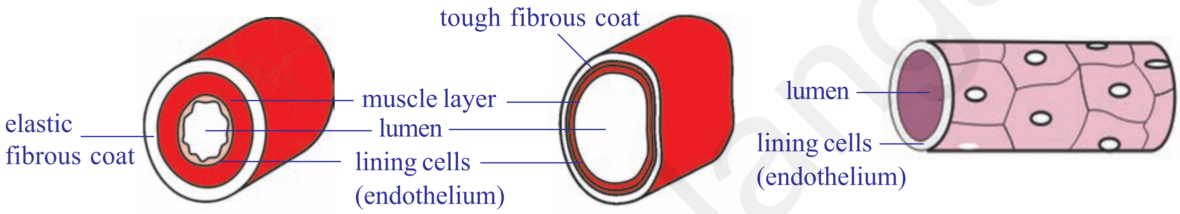


Fig-9(a): T.S. of Artery *Fig-9(b): T.S. of Vein* *Fig-9(c): T.S.of Blood capillary*

Blood capillaries

Blood capillaries are the microscopic vessels made of single layer of cells. They allow diffusion of various substances. The leucocytes (WBC) can squeeze out of the capillary wall. They establish continuity between arteries and veins.

Now fill in the following table.

Table-3

	Structure / Function of blood vessel	Artery	Vein
1.	Thickness of walls(thick / thin)		
2.	Valves (present / absent)		
3.	Capacity to retain shape when blood is absent (can retain/can't retain)		
4.	Direction of blood flow (heart to organs / body organs to heart)		
5.	Pressure in the vessel(low /high)		
6.	Type of blood transported (oxygenated / de-oxygenated)		

- *Discuss the differences between pulmonary artery and pulmonary vein.*

Activity-4

Let us perform the following activities to observe arteries and veins again.

Sit on a table with one leg dangling and the other resting on it so that the back of one knee rests on the knee of the other. After a time you will see and feel the leg which is on top give a series of small movements with each heart beat. If you do it for long you will reduce the blood flow to the leg and so develop ‘pins and needles’. Discuss with your teacher about reasons.

Swing your arm round several times to fill the veins with blood, hold the arm vertically downwards and gently press your finger along a prominent vein-stroking it in the reverse direction to the blood flow, i.e., towards the hand. Can you see the swellings where you have pushed blood against the valves? Discuss with your teacher about reasons.

Think and discuss

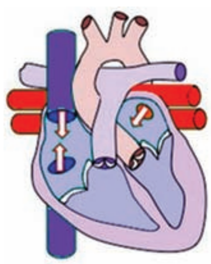
- Artery walls are very strong and elastic. why?
- Why we compare arteries like tree which divides into smaller and smaller branches.
- The lumen size is bigger in vein when compared with artery. Why?

The cardiac cycle

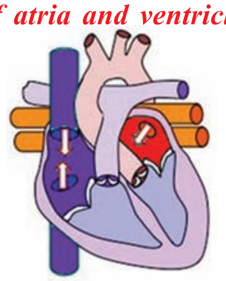
The human heart starts beating around 21st day during the embryonic development (refer reproduction chapter). If it stops beating, it results in the death of a person.

One contraction and one relaxation of atria and ventricles is called one cardiac cycle.

1. We start with the imagination that all the four chambers of the heart are in relaxed state (diastole).
2. Blood from venacava and pulmonary veins enters the right and left atria respectively.
3. Now the atria contract, forcing the blood to enter into the ventricles.
4. When the ventricles are filled with blood they start contracting and atria start relaxing. The aperture between the atria and ventricles is closed by valves. On ventricular contraction due to pressure the blood moves into the aorta and pulmonary artery. When the valves are closed forcibly, we can listen to the first sharp sound of the heart ‘lub’.



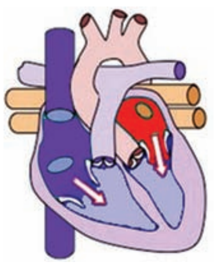
1. Imaginary relaxation of atria and ventricles.



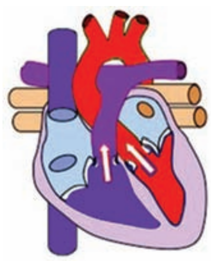
2. Blood flows into atria.

Fig-10(1-2)

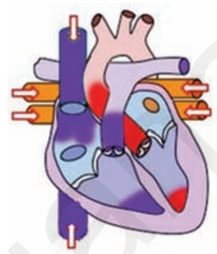
5. When the ventricles start relaxing the pressure in the ventricles is reduced. The blood which has entered the arteries tries to come back into the ventricles. The valves which are present in the blood vessels are closed to prevent backward flow of blood into the ventricles. Now we can listen to a dull sound of the heart 'dub'. The atria are filled up with blood and are ready to pump the blood into the ventricles.



3. Contraction of atria and flow of blood into ventricles.



4. Contraction of ventricles. A.V. Valves closed (Lub) blood flows into arteries.



5. Relaxation of ventricles. The closing of arterial valves (Dub).

Fig-10(1-5): Cardiac cycle

The sequential events in the heart which are cyclically repeated are called cardiac cycle. The cardiac cycle includes an active phase systole and a resting phase the diastole of atria and ventricles.

The time needed for atrial contraction (atrial systole) is 0.11-0.14 seconds. The time needed for ventricular contraction (ventricular systole) is 0.27-0.35 seconds. The time needed for relaxation of both auricles and ventriculars i.e. joint diastole is about 0.4 seconds. The whole process is completed in approximately 0.8 seconds.

Blood is pumped into the blood vessels at regular intervals. The tissues do not receive the blood continuously, but in the form of spurts. When we keep our finger at the wrist, where the artery is passing into the hand we feel the pressure of blood moving in it. This is the 'pulse'. The rate of the pulse will be equal to the number of heartbeats.

Do you know?

Name of the animal	Weight of the body	Weight of the heart	No. of beats/min
Blue whale	1,50,000 kg	750 kg	7
Elephant	3000 kg	12 - 21 kg	46
Man	60-70kg	300 gm	76
Coaltit (Bird)	8 gm	0.15 gm	1200

Single and double circulation

We know that blood flows in the blood vessels. To keep the blood moving the heart pumps it continuously. The blood that is pumped by the heart reaches the body parts and comes back to the heart. But course taken by the blood is not the same in all the animals. Let us observe the fig-11(a) and (b). Start from any point in the fig-11(a) and (b). Move in the direction of arrow. Note down the parts which are in the way in cyclical form.

(Try to identify different parts of the body in both figures.)

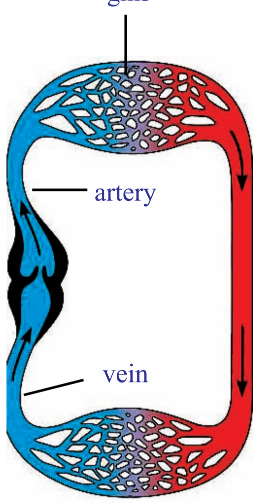


Fig-11(a): Single circulation

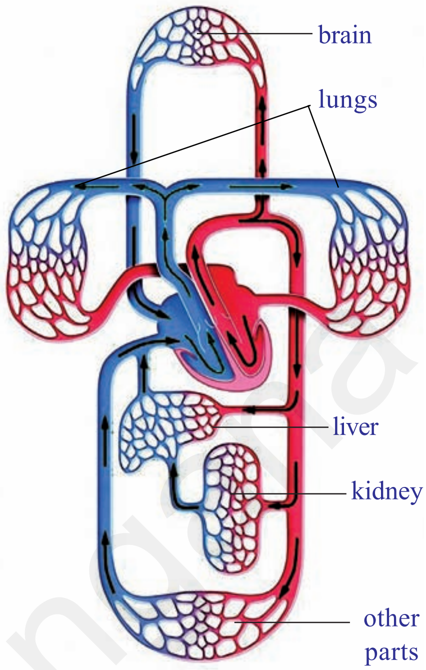


Fig-11(b): Double circulation

Compare the two flow charts and answer the following.

- *How many times did your pointer touch body parts in fig-11(a) and (b)?*
- *How many times did your pointer touch the heart in fig-11(a) and (b)?*
- *How many times did your pointer touch the respiratory organs in fig-11(a) and (b)?*

From your observation it is clear that in fig -11(a) blood flows through heart only once to complete one circulation. This is single circulation.

In Fig-11 (b) blood flows through the heart twice for completing one circulation. This is called double circulation.

Lymphatic system

Have you ever observed what happened to your feet after overnight journey, in sitting position without moving? We feel that our foot wear is little tight. In elders it will be clear that the lower part of the legs will be swollen. This stage is called oedema.

- *Why do our legs swell?*

As blood flows through tissues and through blood capillaries some amount of fluids and certain solid materials are constantly flowing out of them at different junctions. Such materials are collected and sent back into blood circulation.

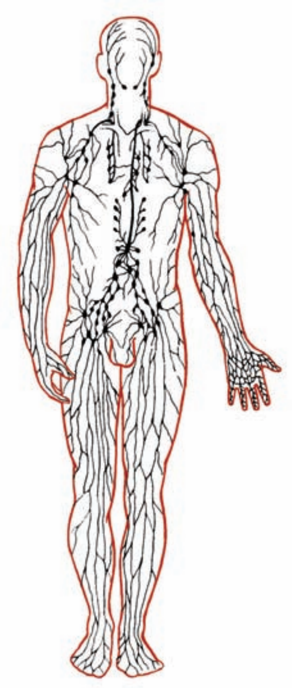


Fig-12: Lymphatic system

We know that blood circulates in the blood vessels, pushed by the heart. From the heart it flows into the arteries and finally into the capillaries. To supply nutrients to the cells (tissues), the liquid portion of the blood with nutrients flows out of the capillaries. This is called tissue fluid.

This fluid flows through a system called as lymphatic system. The fluid mainly returns back into the blood stream.

Lymph is the vital link between blood and tissues by which essential substances pass from blood to cells and excretory products from cells to blood.

Blood is a substance which contains solid and liquid particles. Lymph is the substance that contains blood without solid particles. Tissue fluid is lymph present in the tissues. The liquid portion after formation of blood clot is serum.

The muscles which are attached to the skeleton (skeletal muscles) act as pumps when they contract and help in pushing the lymph flowing in lymphatic vessels and the blood flowing in veins towards the heart.

The valves that are present in the lymphatic vessels and veins stop the reverse flow of blood. We shall read about this as the system of lymph ***circulation*** in detail in higher classes.

Evolution of the transport (circulatory) system

When the unicellular organisms separated themselves from the sea with the formation of the limiting membrane, the problem of transportation arose. The nature has found the solution, by creating a microscopic ocean which has its own currents.

In unicellular organisms like Amoeba the protoplasm shows natural movements. These movements are called Brownian movements, because of which the nutrients and oxygen are distributed throughout the protoplasm equally.

This simplest intracellular transportation system, present in unicellular animals has been retained in multicellular animals including humans. The protoplasm of any cell in our body is mobile and protoplasmic currents exist even in the nerve cells.

The multicellular animals have to develop more complicated system for transportation of materials.

The parazoans like sponges, use sea water for transportation. Since the natural water currents are not reliable, the sponges create their own currents by beating of flagella that are present in their body.

The cnidarians which are better evolved than sponges (e.g. Hydra and jelly fish) have developed blind sac like gastrovascular cavity, which has taken up the function of digestion and transportation of nutrients to each and every cell of the body.

In platyhelminthes (e.g. Fasciola hepatica), the digestive system is highly branched and supplies digested food to all the cells directly. In these animals the excretory system collects wastes from each cell individually. In these organisms most of the body is occupied by digestive and excretory systems.

In animals belonging to Nemathelminthes (round worms), the pseudocoelom has taken up the function of collection and distribution of materials.

The Annelids, the first Eucoelomate animals have developed a pulsatile vessel, to move the fluid and the transporting medium the blood.

The Arthropods have developed a pulsatile organ to pump the blood, the heart. The blood instead of flowing in blood vessels floods the tissues, directly supplying the nutrients to the tissues. Oxygen is directly supplied to the tissues by the respiratory system.

Transportation system which supplies nutrients to the tissues directly is called open type of circulatory system.eg. Arthropods, many molluscs and lower chordates.

The other type of transportation system where the blood takes the responsibility of delivering the materials, which flows in the blood vessels is called closed type of circulatory system. Such type of closed circulatory system is present in annelids echinoderms, cephalopod molluscs (e.g. Octopus) and all the higher chordates.

Do you know?

The human circulatory system can move one ml of blood from heart to a foot and back which is approximately 2 meters, in about 60 seconds.

It would take more than 60 years for the substance to move across this distance by diffusion.

Blood pressure (B.P.)

Movement of blood through the network of vessels requires a great deal of pressure. The pressure provided by the heart is at its highest when the ventricles contract, forcing the blood out of the heart into the arteries. Then there is a drop in the pressure as the ventricles refill with blood for the next beat.

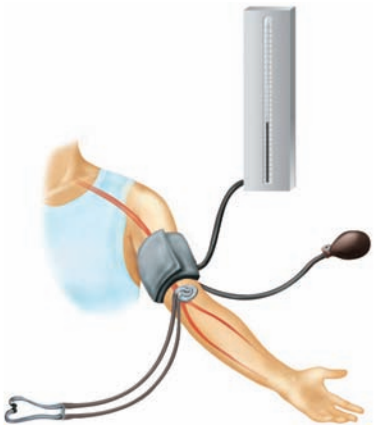


Fig-13: Sphygmomanometer

Blood pressure is the pressure exerted by the blood on the walls of the blood vessels. The pressure is developed due to the contraction of ventricles to push the blood into the vessels. B.P. is usually measured in the upper arm artery as it varies throughout the body. Doctors measure the blood pressure (B.P.) with a device called sphygmomanometer.

There are two readings. One measures the pressure during the time blood is forced out of the ventricles. This is called systolic pressure. For a healthy young adult it will be around 120 mm of Hg. The second reading is taken during the resting period, as the ventricles refill with blood. This is called diastolic pressure. It will be around 80 mm of Hg.

B.P. will change according to the activity in which the person is engaged, such as resting, walking and running. People who have high B.P. during resting period are said to have hypertension.

Coagulation of blood

You know that when you get an injury, the blood flows out of the wound for only a short time. Then the cut is filled with a reddish solid material. This solid is called a blood clot. If blood did not clot, anyone with even a slight wound bleeds profusely.

- When the blood flows out, the platelets release an enzyme called thrombokinase.
- Thrombokinase acts on another substance present in the blood called pro-thrombin converting it into thrombin.
- Thrombin acts on another substance called fibrinogen, that is present in dissolved state converting it into insoluble fibrin.
- The blood cells entangle in the fibrin fibres forming the clot.
- The fibrin fibres are attached to the edges of the wound and pull them together.

The yellowish straw coloured fluid portion after formation of the clot is *serum*.

$$\text{Prothrombrin} \xrightarrow{\text{Thrombrokinase}} \text{Thrombin}$$

$$\text{Fibrinogen} \xrightarrow{\text{Thrombin}} \text{Fibrin}$$

Discuss with your teacher about vitamin K in relation to coagulation of blood.

Normally the blood that oozes from a wound clots in 3-6 minutes. But in some people due to vitamin K deficiency it takes more time. Also due to a genetic disorder blood may not clot after an injury. This type of disorder is called haemophilia. Haemophilia is common disorder in the

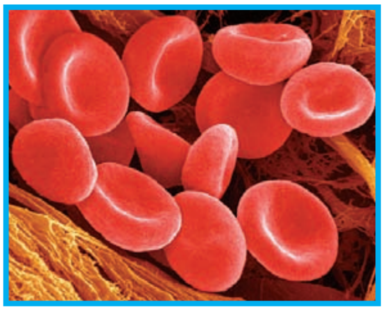


Fig-14(a): Blood in the blood vessel

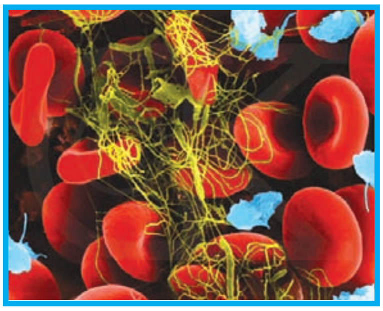


Fig-14(b): Clot formation

children who have born from marriages between very close relatives. Thalassemia an inherited disorder is also related to blood. For more details see annexure.

HOW MATERIALS TRANSPORT WITH IN THE PLANT

There is a vast transport system in continual supply of essential nutrients and oxygen to perform metabolic activities, and to remove excretory substances which are found in each cell of animal body.

Is there anything like that in plants which corresponds to circulatory system?

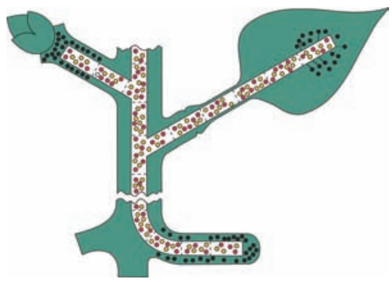


Fig-15: Transportation

In previous classes we studied about Van Helmont's experiments on plants, which get water that contain minerals from soil through their roots. The water absorbed by roots and food prepared by leaves are supplied to the remaining parts of the plant by vascular bundles having xylem and phloem.

In the root, the xylem tissue develops from periphery to center, whereas in the stem it develops from center to periphery in each vascular bundle.

How is water absorbed?

We know that roots absorb water with minerals from soil.

- *What is the mechanism behind this?*
- *Are roots directly in contact with water?*
- *How is water absorbed?*

Activity-5

Absorbing root hairs

To perform this activity, you need to germinate bajra or mustard seeds.

Examine some mustard seedlings which have been grown on wet filter paper. Observe the mass of fine threads coming from the seed by hand lens. These are roots. They have fine microscopic structures called root hairs. These are root hairs through which water enters the plant. Gently squash a portion of the root hair between slide and cover slip in a drop of water and examine under a microscope. Note the thinness of the walls of root hairs. Osmosis plays an important role in movement of water from root hairs to xylem.

Every living cell acts as an osmotic system, the cytoplasm lining of the cell wall acts as the semipermeable membrane. Observe the following figure, notice how do roots penetrate into soil? You will find that the root hairs grow out into the spaces between the soil particles and that the hairs are surrounded by moisture.

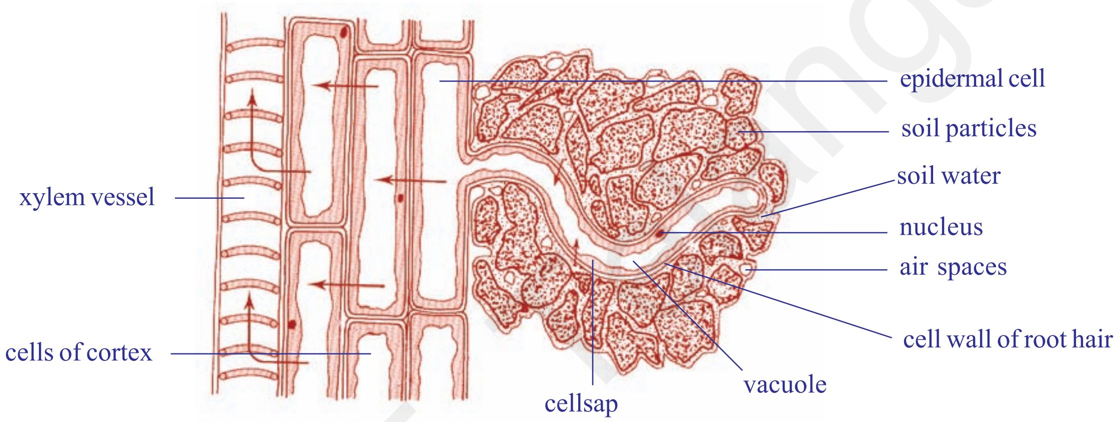


Fig-16: L.S of root showing relationship of root hair and soil water

Note: *In Fig-16 Arrow marks shows the direction of flow of water.*

The soil water is an extremely dilute solution of salts. Soil water concentration is more dilute than that of the cell sap in the root hair; therefore water will pass into the vacuole of the root hair by osmosis. Recall the process of osmosis that you have learnt in the chapter "moving of substances through plasma membrane" in class IX. The entry of water dilutes the contents of the root hair vacuole so that it becomes more dilute than it's neighbouring cell.

So, water passes into the neighbouring cell which in turn becomes diluted, finally water enters the xylem vessels. As there are vast number of root hairs and root cells involved, a pressure in the xylem vessels develops which forces the water upwards. This total pressure is known as root pressure. Root pressure is not the main cause of movement of water in xylem but it is certainly one of the factor. The other factors are also there. You will learn about those reasons in higher classes in detail.

Activity-6

What is root pressure

Take a regularly watered potted plant and cut the stem portion 1 cm above the ground level. Then connect a glass tube by means of strong rubber tubing as shown in the figure. The size of glass tube should be equal to the size of the stem. Take care while joining tube and stem being bound tightly, so that water cannot escape from the tube. Now, pour some water in the glass tube until water level can be seen above the rubber tube. Mark the level of water (M_1) in the tube. Keep your arrangement aside for 2 to 3 hours. Then observe and mark the water level (M_2) in the tube.

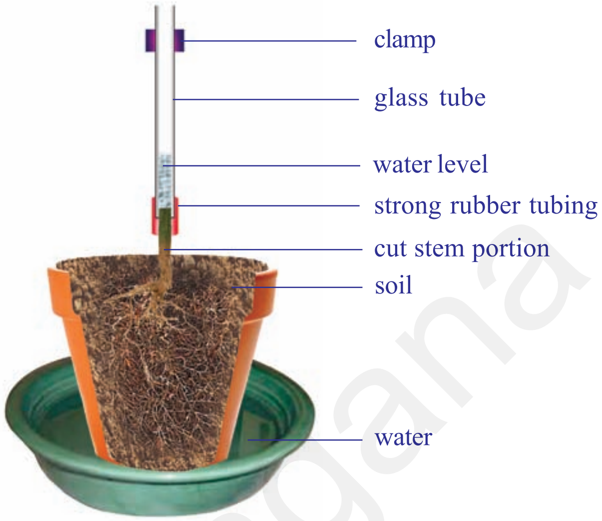


Fig-17: Root pressure

- *Is there any increase in the water level?*
- *What is the role of xylem in this action?*

The difference between M_2 and M-$_1$ indicates the level of water raised in the stem. Because of the root pressure, the water level increases in the tube.

The mechanism of water movement in plants

We have seen that there is a push from below due to root pressure on the columns of water in the xylem vessels, but this is seldom high and in some seasons it is nil. How does the water reach 180 metres high to the top of a tree like a eucalyptus?

Let us recall the activity that you performed in lower classes. Why inner sides of polythene cover become moist? Where do these water droplets or water vapour come from?

We know that this type of evaporation of water through leaves is called 'transpiration'. Water evaporates through stomata of leaves and lenticels of stem.

When the leaves transpire, there is a pulling effect on the continuous columns of water in the xylem vessels. The top ends of these vessels are surrounded by the leaf's mesophyll cells which contain cell sap, so the water is continuous from the xylem vessels to the walls of the

Fig-18: Transpiration

mesophyll cells from which it evaporates into the air spaces causing the pull. The water column does not break because of its continuous molecular attraction. This is a property of water you demonstrate every time you drink through a straw.

Now we have a picture of the water-conducting system of a tree. Water is absorbed by osmosis from the soil by the root hairs. This is passed into the xylem vessels which form a continuous system of tubes through root and stem into the leaves. Here the water evaporates and releases into the atmosphere. The evaporation creates the main pull of water above root pressure which gives a variable and minor push from below. This results in a continuous column of moving water, the 'transpiration stream'.

Is there any relation between transpiration and rain fall?

The amount of water passing through a plant is often considerable. For example, an oak tree can transpire as much as 900 liters of water per day. It follows therefore that areas of forest significantly affect the degree of saturation of the air above them, so that when air currents bring air which is already nearly saturated to a forest area, it becomes fully saturated and comes down as rain; this is why forest areas often have a higher rainfall than areas nearby.

Do you know?

How much water is transpired by plants? Each fully grown maize plant transpires 15 liters per week. One acre of maize may transpires more than 13,25,000 liters of water in a hundred day growing season. A big mango tree will transpire from 750 to more than 3500 liters of water per day during growing season.

Transport of mineral salts

You know that mineral salts are necessary for plant nutrition (micro and macro nutrients) and that they are obtained from the soil solution through the root hairs. The salts are in the form of electrically charged ions. Sodium chloride (NaCl) is in the form of Na^+ and Cl^-, and Magnesium Sulphate ($MgSO_4$) occurs as Mg^{2+} and SO_4^{2-}. But, they are not absorbed into the root hairs by the simple process of diffusion, but it involves the use of energy by the cytoplasm which will be discussed in later classes. Once ions are absorbed, the ions travel along with water in the xylem vessels and pass to the growing points of the plants where they are used for growth purpose. They may also pass laterally from xylem to phloem. Thus, mineral salts are one of the natural factors in plant growth phenomena.

Transport of materials in plants

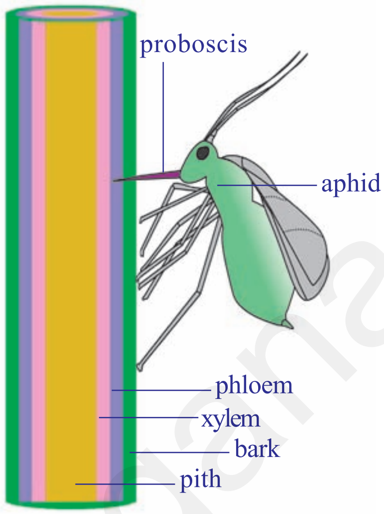


Fig-19: Aphid extracting food material from plant

Food such as sugar is synthesised in the green parts of plants, mainly the leaves, but this food has to be transported to all the living cells, especially to actively growing cells and the cells which stores food.

The veins of a leaf consist of xylem and phloem, and these tissues are continuous with the stem. The following experiments provide evidence that food is transported in the phloem cells.

Phloem sieve tubes are extremely small and the analysis of their contents is not easy. Biologists studied about food transportation in plants with the help of aphids (greenfly). When you see aphids clustering round the young stems of plants as they feed on the plant juices. To obtain this juice an aphid pierces the plant tissues with its long needle like organ "proboscis". It can be shown when a feeding aphid is killed and the stem carefully sectioned, the proboscis only penetrates upto a phloem sieve tube. This proboscis also provides a ready-made means of obtaining the juice for analysis! The experiment can be done in this way. An aphid is killed while in the act of feeding and the body is then carefully cut away, leaving the hollow proboscis still inserted into the phloem. It is found that because the contents of the phloem sieve tubes are under slight pressure the fluid slowly exudes from the cut end of the proboscis in the form of drops; these drops are then collected and analysed. The fluid is found to contain sugars and amino acids.

Not surprisingly, aphids absorb so much sugar from the phloem that they cannot assimilate all of it and it excretes out of the body as a sticky syrup called honey dew. Leaves which have been attacked by aphids often feel sticky as a result of honey dew.

You may notice that sometimes barks of the tree damaged more than a half, even then the tree is alive. How is this possible?

Further experiments to illustrate the conduction of sugars by the phloem have been done by removing a ring of bark from a shoot to expose the wood. Remove all tissues from the center outwards, including the phloem. After a

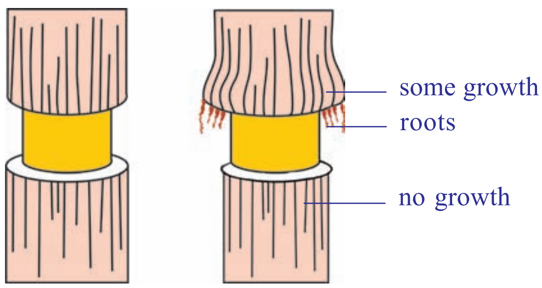


Fig-20: Removing ring of bark (Ringing experiment) few days, when the tissues above and below the ring were analyzed it was shown that food had accumulated above the ring, but was not present below it. If it is left for some more time, the stem increases in thickness immediately above the ring, but no growth occurred below it. So, any damage to the phloem all around the stem will prevent the food from passing down to the roots and the tree will eventually die. This is a fact of great economic importance because certain mammals scratching the bark of trees to get the food stored in the phloem, especially during hard winters when food is scarce. Voles (rat like rodents) do this to young saplings at ground level and rabbits can do much damage to older ones. Foresters find it economically worthwhile to enclose new plantations with wire netting to prevent rabbits from entering.

Key words

Circulation, Auricles, Ventricles, Pulse, Artery, Vein, Stethoscope, Aorta, Capillary, Systole, Diastole, Cardiac cycle, Blood pressure, Lymph, Single circulation, Double circulation, Coagulation of blood, Sphygmomanometer, Prothrombin, Thrombin, Fibrinogen, Fibrin, Root hair, Radical, Root pressure, Plant nutrients, Xylem, Phloem, Vascular bundles.

What we have learnt

- The pulse rate is equal to heart beat. We can count the heart beat without the aid of any instrument.
- Rene Laennec discovered the first stethoscope.
- The heart is covered with two pericardial membranes filled with pericardial fluid which protects it from shocks.
- Rigid blood vessels attached to the heart are called arteries. The two rigid blood vessels are arteries which supplies blood to body parts and lungs.
- The less rigid vessels are veins, which brings blood from body parts.
- Heart has four chambers, two upper atria and two lower ventricles.
- Atrium and ventricle of the same side are connected by atrio ventricular aperture.
- Atria are separated from each other by interatrial septum, ventricles by interventricular septum.
- The atrioventricular apertures are guarded by valves. There are valves in the aorta and pulmonary artery also.

- The right side of heart receives blood from body and sends to lungs.
- The left side of the heart receives blood from lungs and send it to body parts.
- The arteries carry oxygenated blood except pulmonary artery. The veins carry deoxygenated blood except pulmonary veins.
- One contraction and relaxation of heart is called cardiac cycle.
- If the blood goes to heart only once before it reaches all the body parts. It is called single circulation. If it goes twice it is called double circulation.
- Vitamin K deficiency leads to delayed coagulation of blood.
- Plants absorb soil water through roots by the process of osmosis.
- Water travels through xylem vessels and food material travels through phloem tissues.
- There is a relation between tranportation and transpiration in plants.
- Biologists studied about phloem tubes with the help of aphids.

Improve your learning

N8I8J3

1. What is transport system? How does this help the organism?(AS1)
2. What is the relationship between blood and plasma?(AS1)
3. Which type of blood vessels carry blood away from the heart?(AS1)
4. What are the three main types of blood vessels in the body?(AS1)
5. Which is the largest artery in the body? Why is it the largest ?(AS1)
6. Which blood vessel carries blood for oxidation?(AS1)
7. Name the structures which are present in veins and lymph ducts and absent in arteries.(AS1)
8. What is the use of platelets?(AS1)
9. Write differences between (AS1)

   a) systole - diastole   b) veins - arteries   c) xylem - phloem
10. Explain the way how the plants absorb water from soil ?(AS1)
11. What is root pressure? How it is useful to the plant?(AS1)
12. Phloem is a food source for some animals. How can you justify this statement?(AS1)
13. Read the given para and name the parts of the heart.(AS1)

We have observed that the heart is divided into four chambers by muscular structure. Any structure that divides two chambers is known as ***septum***. Now let us try to name the septa present in the heart.

a) The septum that divides the two atria can be named as ______________

b) The septum that divides the two ventricles can be named as____________.

c) The septum that divides the atrium and ventricle can be named as_________.

The holes that connect two chambers are called ***apertures***. Let us try to name the apertures which connect the atria and ventricles.

d) The aperture that connects the right atrium and right ventricle can be named as________.

e) The aperture that connects the left atrium and left ventricle can be named ____________.

Any structure that closes an aperture, and allows one way movement of materials is called as ***valve***. Now let us name the valves that are present in the chambers of the heart.

f) The valve that is present between left atrium and left ventricle can be named as____________.

g) The valve that is present between right atrium and right ventricle can be named as ___________.

14. If the valves in veins of the legs fail to stop the flow of blood what could be the consequences of this failure?(AS2)
15. What would happen if transpinration doesn't take place in plants?(AS2)
16. John made a stethoscope using a paper cup and plastic tube. Write down the procedure he followed. (AS3)
17. How did scientists prove that the food is transported through the phloem?(AS3)
18. What is your inference about experiments with aphids?(AS3)
19. Collect information about blood pressure of your school teachers or neighbours. Prepare a report on their health problems due to changes in blood pressure. (AS4)
20. Draw a schematic diagram to explain single and double circulation. Write differences between them.(AS5)
21. Prepare a block diagram showing water absorption by roots to transpiration by leaves . (AS5)
22. What can circulatory system in man be compared with? (AS6)
23. What is Haemophilia? (AS1)
24. Prepare a cartoon on heart beating? (AS7)
25. After reading this lesson what precautions would you suggest to your elders about edima.(AS7)

Choose the correct answer

1. The term cardiac refers to which organ in the body? ( )

   A) heart B) vein C) lymph D) capillary

2. In which chamber of the human heart the blood is low in oxygen? ( )

   A) right atrium B) right ventricle C) left atrium D) A and B

3. Which structures of the heart control the flow of the blood? ( )

   A) arteries B) veins C) valves D) capillaries

4. Which of the following statement is wrong? ( )

   A) Serum is the liquid portion formed after blood clotting.

   B) Lymph is the link between blood and tissues.

   C) The xylem and phloem transport water and food in plants.

   D) In insects closed type of circulatory system is seen.

5. An aphid pierces its proboscis into the .... to get plant juices ( )

   A) Xylem B) phloem C) cambium D) vascular bundle

Annexure-I

The rhesus factor

There is another antigen of red blood cells which is present in 85% of the people of Britain, this is known as the Rhesus factor, as it was first discovered in Rhesus monkeys. People who have this are said to be Rhesus positive (Rh+). Those who do not have this factor are termed Rhesus negative (Rh-). Normally they do not carry an antibody to this factor in their plasma. However, if Rh+ blood is transfused into the blood of a Rh- person, against Rh-factors will be formed and these are capable of destroying Rh+ red cells. Under certain circumstances this is a potential hazard for babies.

If a Rh+ man marries a Rh- woman, some of the children are likely to be Rh+. At birth there is always some mixing of blood between the circulation of mother and baby and this may occasionally happen during pregnancy. So, if a child is Rh+ some of its blood will leak into its mother's circulation and cause antibodies to form in her blood. If the mother has more children, not all will necessarily be Rh+, but if they are, the amount of antibodies in her blood often increases with each pregnancy, and in some instances the antibodies in her blood may pass into the baby's blood in sufficient quantities to produce very serious anaemia and even death. Fortunately these cases are not frequent, and when they do occur, the baby is given a complete transfusion soon after birth so that the baby's blood is replaced by blood containing no antibodies to the rhesus factor. It is now possible for this transfusion to be carrried out before birth. Another recently developed technique is for the mother to be given an injection shortly after the birth of her first child which prevents the Rh+ cells from stimulating the production of the harmful antibody.

Annexure-II

Thalassemia

Thalassemia is a group of inherited blood disorders characterized by mild to severe anaemia caused by haemoglobin deficiency in the red blood cells. In individuals with thalassemia, the production of the oxygen carrying blood pigment haemoglobin is abnormally low. There are two main types of thalassemia: alpha thalassemia and beta thalassemia. In each variant a different part of the haemoglobin protein is defective. Individuals with mild thalassemia may have symptoms, such as anaemia, enlarged liver and spleen, increased susceptibility to infections, slow growth, thin and brittle bones, and heart failure.

Facts about Thalassemia

- Thalassemia is a serious Inherited Blood Disorder.
- 4.5% of world population (250 million) suffering with thalassemia minor.
- About 35 million Indians are carriers of the abnormal Gene for Thalassemia.
- It is estimated that about 1,00,000 infants are born with major Haemoglobinopathies every year in the world.
- 10,000 – 12,000 Thalassemic children are born every year in our country.
- Survival depends upon repeated blood transfusion and costly medicines.
- Thalassemia can be prevented by awareness, pre marital or pre conceptual screening followed by antenatal diagnosis is required.

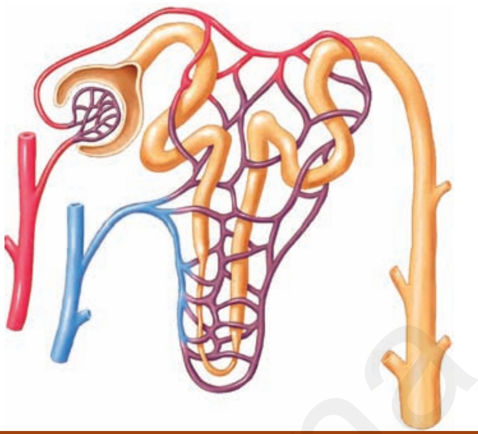
Treatment

Thalassemia major should be diagnosed as early as possible in order to prevent growth restriction, fragile bones and infections in the first year of life. The infant's haemoglobin levels and development should therefore be monitored closely. If Hb is less than 70% or the child shows signs of poor growth and development. Regular transfusion is the treatment of choice. According to the WHO, the aim of this treatment is to retain a median haemoglobin value of 115–120 grams per liter. This can usually be achieved by carrying out transfusions of concentrated red blood cells at intervals of every three to four weeks.

Today thalassemia major can be cured by stem cell transplantation. A prerequisite is usually that the affected individual who has siblings with identical tissue type (HLA type) a transplantation of blood stem cells referred to as a "bone marrow transplant", can be carried out.

Chapter 4

W3Q2B5



Excretion

There is no factory which can manufacture a product without generating any waste. This is true to our body which is a living cellular factory and for other organisms as well. Wastes are generated at regular intervals from the bodies of many organisms. This raise questions like.

- *Where are the wastes produced?*
- *How are they produced?*
- *What are the substances present in them?*
- *Does the composition vary in the same organism in different situations?*

Let us understand such kind of questions.

Living beings need energy for their survival and to perform activities either building up of body material (anabolism) or its breakdown (catabolism), collectively called metabolic activities. Organisms use different substances for metabolic activities. Different products are generated as a result of these metabolic activities. Can you name different products generated by the following life processes?

Table-1

Life processes	Products
Photosynthesis	
Respiration	
Digestion	

- *What products would the organism be able to take up for other activities?*
- *What are the products which would cause harm to the body, if they are not removed?*
- *What happens if harmful products are not removed from our body every day?*

We have already learnt that different kinds of materials are produced out of various metabolic activities. Some of these may be harmful for the organism are removed from their body or packed and stored in some other forms. These are all wastes produced in the body of an organism. We have already discussed how organisms get rid of gaseous wastes generated during photosynthesis or respiration. Other metabolic activities generate nitrogenous wastes that have to be removed along with salts, excess water and several other materials. Excretion is the term coined for all the biological process involved in separation and removal of wastes or non useful products from the body. (In latin ex means out, crenere means shift.) Now let us study how excretion takes place in human being.

Excretion in Human Beings

A number of reactions take place during various metabolic activities. Many useful substances and energy are produced. At the same time many other things happen such as, toxic wastes may be produced, water content may increase, ionic balance (homeostasis) in the body may be disturbed. The waste products include carbon dioxide, water, nitrogenous compounds like ammonia, urea, uric acid, bile pigments, excess salts etc. The most poisonous of all waste products of metabolism is ammonia.

Where are these waste materials produced? How does the body manage them. Is there a way to detect their presence in our body?

Now let us observe the test reports of Blood and Urine of a person given in tables-2 and 3. Find out the components present in both Blood and Urine. (For 24 hours urine test urine is collected for 24 hours. From that, 100-150 ml sample will be tested.) observe the reports in the next page and answer the following questions.

- *What are the substances present in blood?*
- *What are the substances present in urine?*
- *What are the substances present both in blood and urine?*
- *Which substances are present above the normal limits both in the blood and urine?*
- *What do you think a reading above normal limits indicates?*

Table-2: DEPARTMENT OF BIOCHEMISTRY

REPORT ON PLASMA/SERUM (BLOOD) ANALYSIS OF A PATIENT

TEST NAME	RESULT	UNITS	RANGE
GLUCOSE FASTING	82	mg/dl	60-100
SODIUM	137	mmoles/L	135-145
POTASSIUM	4.10	mmoles/L	3.5-5.0
CHLORIDES	101	mmoles/L	95-106
UREA	29	mg/dl	15-40
CREATININE	2.8.	mg/dl	0.6-1.5
URIC ACID	7.50	mg/dl	3.0-5.0
TOTAL CHOLESTEROL	221	mg/dl	150-200
TRIGLYCERIDES	167	mg/dl	60-200
CALCIUM	9.40	mg/dl	8.0-10.5
PHOSPHORUS	4.50	mg/dl	3-4.5
BILIRUBIN (TOTAL)	0.70	mg/dl	0.1-0.8
TOTAL PROTEINS	7.20	g/dl	6.0-7.5
ALBUMIN	4.60	g/dl	3.0-5.0

Table-3: DEPARTMENT OF BIOCHEMISTRY

REPORT ON URINE ANALYSIS OF THE SAME PATIENT

TEST/METHOD	RESULT	UNITS	RANGE
24 hrs.Protein	90	mg/day	<100 mg
24 hrs Creatinine	2.7	mg/day	1-2
24 hrs.Calcium	305	mg/day	Up to 200
24hrs.phosphorous	0.8	mg/day	upto 1g
24hrs.uric Acid	800	mg/day	upto 600
ELECTROLYTES :			
Sodium	140	mmoles/L	125-250
potassium	50	mmoles/L	25-100
Osmolality (calculated)	180	mmoles/L	100-600
Glucose	65	mg/dl	50-80
Chlorides	128	mmoles/L	120-130
Urea	35	gm/day	20-30

m moles / L means millimoles per litre, mg/dl means milligram per deci litre.

- *What are the materials needed to be removed from our body?*
- *Where are these materials removed from?*
- *Why do you think the body must remove waste substances?*

Studying the structure and function of our excretory system will help us to understand this better.

Excretory System in Human being

In human beings excretion mainly occurs through a urinary or excretory system it consists of a pair of kidneys, a pair of ureters, urinary bladder and urethra, as shown in the fig-4. Now let us observe external and internal features of a kidney in goat / sheep, which is similar to Human kidney in function.

Aim: Studying the external and internal features of a kidney

Lab Activity

Materials required: Freshly collected specimen of sheep/goat's kidney from the butcher or 3D Model of a kidney, sharp blade/scalpel, tray and a jug of water.

Procedure for observation: Before bringing the kidney to the lab wash it thoroughly so that, blood is completely drained from it. Put the kidney in the tray and observe it carefully. Note your observations in the observation book. With the help of sharp blade take a longitudinal section here you are advised to do this activity under the guidance of your teacher and observe the internal structure.

Fig-1: Kidney of goat

Draw what you have observed and compare it with Fig-1 and 2.

- *What is the shape of kidney?*
- *What is the colour of kidney?*
- *Do you find any attachments on upper portion of kidney?*
- *Is the Internal structures similar to fig-2*
- *What is the colour of the outer part in L.S of kidney?*
- *In L.S of kidney where do you find dark brown colour portion?*
- *How many tubes are coming out from kidney fissure?*

Fig-2: LS of Kidney of goat

Don't forget to wash your hands with antibacterial lotion after completing dissection.

Now let us know the structure of human excretory system and its functions.

1. Kidneys

In Human beings there is a pair of bean shaped, reddish brown structures in the abdominal cavity attached to dorsal body wall (Fig-3) one on either side of the back bone. These are kidneys. The right kidney is placed slightly lower than the left kidney. Think why it is so?

The size of the kidney is 10 cm in length, 5-6 cm in breadth, and 4 cm in thickness. Each kidney is convex on the outer side and concave on the inner side. The position of the right kidney is lower than the left kidney due to the presence of liver above.

Let us recall the last question in your lab activity. The inner side of each kidney has a fissure or hilum for the entry of a ***renal artery***, exit of a ***renal vein*** and an ***ureter***. Renal artery brings oxygenated blood loaded with waste products and renal vein carries deoxygenated blood. The waste products generated in various organs of the body are filtered and removed by the kidneys.

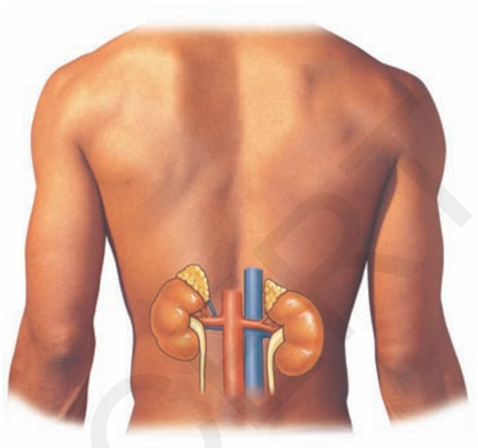


Fig-3: Location of kidneys

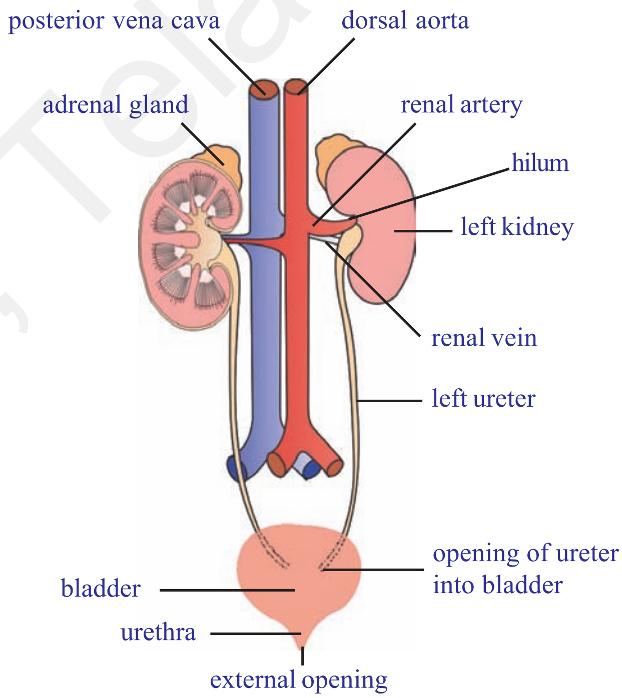


Fig-4: Excretory system

Internal structure of the kidney:

Let us observe L.S of the kidney to know more about internal structure. It shows two distinct regions. Dark coloured outer zone called the cortex and pale inner zone called medulla. Each kidney is made up of approximately more than one million (1.3 to 1.8 million) microscopic and thin tubular functional units called nephrons or uriniferous tubules.

Structure of nephron

Each nephron has the following parts.

Malpighian body: It consists of a blind cup shaped broader end of nephron called Bowman's capsule and bunch of fine blood capillaries called glomerulus. The afferent arteriole (arteriole is the finer branch of an artery) enters into the Bowman's capsule and divides into a network of capillaries and comes out of the capsule as efferent arteriole with lesser diameter. The bunch of capillaries is called glomerulus.

- *Think why the diameter of the efferent arteriole is smaller than that of afferent arteriole?*

Because of the narrower outlet (efferent arteriole) pressure exerts in the glomerulus.It functions as a filtration unit. Bowman's capsule which accommodates one glomerulus,is lined by a single layer of squamous epithelial cells called podocyte cells. There are fine pores between podocyte cells to allow passage of materials filtered out of glomerulus.

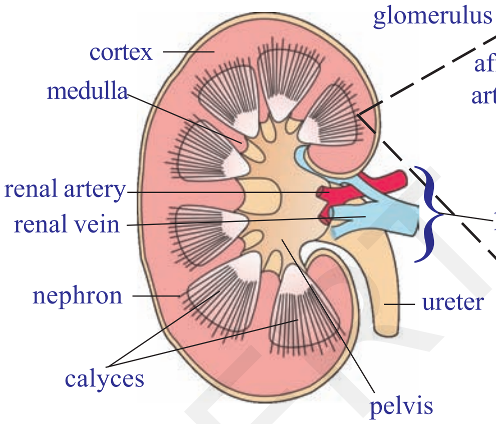


Fig-5: Internal structure of kidney

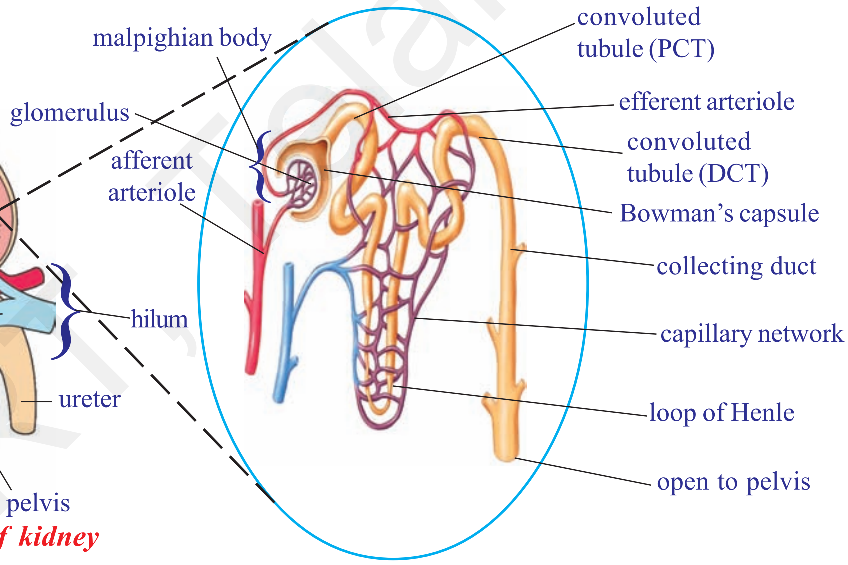


Fig-6: Structure of a nephron

Renal tubule: It has three parts. First or proximal convoluted tubule (PCT), loop of Henle, which is U shaped, second or distal convoluted tubule (DCT).

Distal convoluted tubules open into a collecting tube. Collecting tubules form pyramids and calyces which open into the pelvis. Pelvis leads into the ureter. All the parts of the renal tubule are surrounded by a network of peritubular (around tube) capillaries formed from efferent arteriole. The peritubular capillaries join to form renal venule, which joins the other venules to form finally the renal vein.

- *Why the nephron is considered to be the structural and functional unit of the kidney?*

Mechanism of urine formation

Formation of urine involves different stages. They are :

1. Glomerular filtration
2. Tubular Re-absorption
3. Tubular secretion and
4. Formation of concentrated urine.

1. Glomerular filtration : The which blood flows into glomerulus, it is filtered under high pressure. As a result waste materials along with some water and useful substances are filtered out. They enter into the Bowman's capsule. This is called glomular filtration. This filterate is also called as 'primary urine'.

2. Tubular Re-absorption : Primary urine which is in glomerulus is almost equal to blood in chemical composition except the presence of blood cells. Peritubular capillaries present around the Henle's loop reabsorb essential substances and excess water present in the primary urine.

3. Tubular secretion : After reabsorption of essential substances and water urine travels through the loop of Henle. From peritubular capillaries, present around the loop of Henle, waste materials left unfiltered in the blood during glomerulus filteration are secreted into the loop of Henle.

4. Formation of concentrated Urine : Urine that reaches into the collecting tubule, from the loop of Henle is further concentrated in the presence of hormone vasopressin. Deficiency of vasopressin causes excessive dilute urination called 'diabetes insipidus'.

Do you know?

After the age of 40 years the number of functioning nephrons usually decreases by about 10% in every 10 years.

- *Why more urine is produced in winter?*
- *What happens if reabsorption of water does not takes place?*

Now let us discuss remaining parts of excretory system.

2. Ureters

There are a pair of whitish, narrow distensible and muscular tubes of 30cm length. Each ureter arises from hilus of the kidney. It moves downward and obliquely opens into the urinary bladder. Ureter carries urine from the kidney to the urinary bladder. The movement of urine in the ureter is through peristalysis.

3. Urinary bladder

It is a median, pear shaped and distensible sac. It is situated in the pelvic region on the ventral side of the rectum in the abdomen. It stores urine brought by two ureters. The storage capacity of urinary bladder is 300 - 800ml.

4. Urethra

It is a tube that takes urine from urinary bladder to outside. The opening of urinary bladder into urethra is guarded by a ring of muscles or sphincter. Urethra is 4 cm long in females open to vestibule and in males it is about 20cm long. Its opening is separate in females but is in common with the reproductive tract in males (urino-genital duct).

Micturition

The urine is temporarily stored in the bladder. There are two sets of circular sphincter muscles in the bladder. When the bladder is filling up both these sets of muscles are constricted, so the exit is closed. However as the pressure of the urine increases the walls of the bladder are stretched and this triggers off an automatic reflex action which causes the upper sphincter to relax. But the lower sphincter is under the control of brain. So urine can still be retained until this muscle is relaxed too. Control of urination is not possessed by very young children but is gradually learned.

Urge for micturition occurs when urinary bladder is filled with 300 - 400 ml of urine. The stretched bladder stimulates nerve endings to develop the reflex. However, urine can be retained in the urinary bladder till it gets filled up to the maximum capacity of 700 - 800 ml. At this time the urge becomes painful and leads to voluntary micturition (release of urine from urinary bladder). Total amount of urine excreted per day is about 1.6-1.8 litres. Its quantity increases with larger intake of fluids like water, fruit juices and decreases with lesser intake.

Think and discuss

- Do cells need excretion?
- Why are we advised to take sufficient water?
- Why do some children pass urine during sleep at night until 15 or 16 years of age?

Composition of urine

It is a transparent fluid produced by urinary system. Composition of normal urine varies considerably depending on several factors for instance taking a protein rich diet will result in more formation of urea in the urine. This is because the proteins get de-aminated in the liver with subsequent formation of urea. Even sugar can appear in a normal person after a heavy intake. If other conditions are constant, a large intake of liquids or water - rich food increases the volume of water in the blood, hence more urine is excreted.

Urine contains 96% of water 2.5% of organic substances (urea, uric acid, creatine, creatinine, water soluble vitamins, hormones, and oxalates etc) and 1.5% of inorganic solutes (sodium, chloride, phosphate, sulphate, magnesium, calcium, iodine). It is acidic (pH=6.0)in the beginning but becomes alkaline on standing due to decomposition of urea to form ammonia.

- *What happens if both kidneys fail completely?*

Complete and irreversible kidney failure is called End Stage Renal Disease (ESRD). If kidneys stop working completely, our body is filled with extra water and waste products. This condition is called uremia. Our hands or feet may swell. You feel tired and weak because your body needs clean blood to function properly. Is there any solution to this problem? Let us know about artificial kidney.

Dialysis Machine (Artificial kidney)

Kidneys are vital organs for survival. Several factors like infections, injury, very high blood pressure, very high blood sugar or restricted blood flow to kidneys. This leads to accumulation of poisonous wastes in the body and leads to death. Dialysis machine is used to filter the blood of a person when both kidneys are damaged. The process is called 'haemodialysis'. In this process blood is taken out from the main artery, mixed with an anticoagulant, such as heparin, and then pumped into the apparatus called dialyzer. In this apparatus blood flows through channels or tubes which are madeup of cellophane. These tubes are embedded in the dialyzing fluid. The membrane separates the blood flowing inside the tube and dialyzing fluid (dialysis), which has the

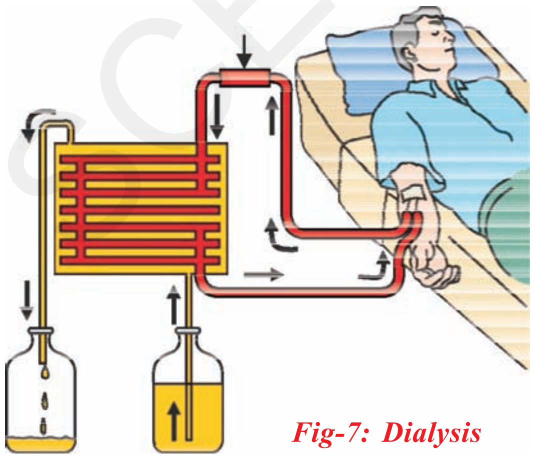


Fig-7: Dialysis

same composition as that of plasma, except the nitrogenous wastes.

As nitrogenous wastes are absent in dialyzing fluids, these substances from the blood move out freely, there by cleaning the blood of its wastes. This process is called dialysis. This is similar to function of the kidney but is different as there is no reabsorption involved .The cleaned blood is pumped back to the body through a vein after adding anti-coagulant (heparin). Each dialysis session lasts for 3 to 6 hours. This method has been using for thousands of uremic / kidney failure patients all over the world.

- *Is there any long term solution for kidney failure patients?*

Kidney transplantation

The best long term solution for kidney failure (acute renal failure) is Kidney transplantation. A functioning kidney is used in transplantation from a donor preferably a close relative. The kidney that is received by a recipient must be a good match to his body, to minimize the chances of rejection of transplanted kidney by the immune system of the recipient. Modern clinical procedures have increased the success rate of such complicated technique.

- *Where is the transplanted kidney fixed in the body of a kidney failure patient?*
- *What about the failed kidneys?*
- *Can donor lead normal life with a single kidney without any complications?*

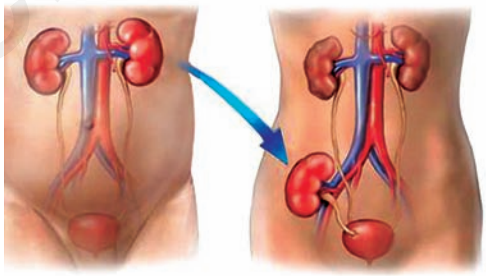


Fig-8: Kidney transplantation

Now a days the process of organ donation helps a lot for kidney failure patients. Organs are collected from brain dead patients, then transplanted to the recepient. To know more about organ donation see in annexure.

Other pathways of excretion (*accessory excretory organs*)

You have learnt about kidney, chief excretory organ of our body.

- *What are the other excretory organs of human body?*

Lungs, skin, liver and large intestine have their own specific functions but carry out excretion as a secondary function.

Lungs: In respiratory process lungs remove carbon dioxide and water.

Skin: It consists of large number of sweat glands

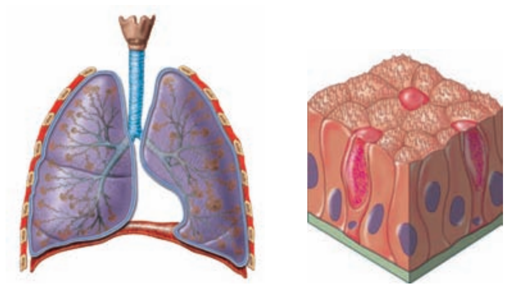


Fig-9 : Lung, Skin

richly supplied with blood capillaries, from which they extract sweat and some metabolic wastes. Since the skin sends out plenty of water and small amount of salts, it serves as an excretory organ. Sebaceous glands in skin eliminate sebum which contains waxes, sterols, hydrocarbons and fatty acids.

- *Collect information on sebum and prepare a news bulletin, display it on bulletin board?*
- *People in cold countries get very less/no sweat. What changes occur in their skin and in other excretory organs?*

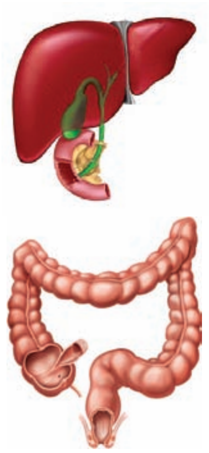


Fig-10: Liver, intestine

Liver: It produces bile pigments (bilirubin ,biliverdin and urochrome) which are metabolic wastes of haemoglobin of dead R.B.Cs. The life span of RBC is 120 days. They are destroyed in the liver. Urochrome is eliminated through urine. Biliverdin and bilirubin are excreted through bile along with cholesterol and derivatives of steroid hormones, extra drug, vitamins and alkaline salts. Liver is involved in urea formation.

Large intestine: Excess salts of calcium, magnesium and iron are excreted by epithelial cells of colon (large intestine) for elimination along with the faeces.

Small amount of nitrogenous wastes are also eliminated through saliva and tears.

Excretion in other organisms

Different organisms use varied strategies in excretion. Specific excretory organs are absent in unicellular organisms. These organisms remove waste products by simple diffusion from the body surface into the surrounding water. Fresh water organisms like *Amoeba, Paramoecium* possess osmoregulatory organelle called contractile vacuole. It collects water and waste from the body, swells up, reaches the surface and bursts to release its content to outside. The main excretion takes place through body surface (Diffusion).

Table-4

Name of the phylum/ organism	Excretory system/ organ
Protozoa	Simple diffusion from the body surface in to the surrounding water
Porifera and coelenterates	Water bathes almost all their cells
Platyhelminthes	Flame cells
Nematoda	Renette cells
Annelids	Nephridia
Arthropoda	Green glands, Malpighian tubules
Mollusca	Meta nephridia
Echinodermata	Water vascular system
Reptiles, Aves and Mammals	Kidneys

Multicellular organisms possess different excretory organs for removal of waste materials from the body. Structural and functional complexity of excretory organs increases from sponges to humans. Sponges and Coelenterates do not have specific excretory organs as water bathes almost all their cells. Excretory structures appear for the first time in Flatworms (Platyhelminthes) are known as flame cells.

Now let us see how this vital process takes place in plants

Excretion and release of substances in plants

Do plants excrete like animals?

We are amazed to answer such type of questions. You are aware that a variety of end products are formed during metabolism and these nitrogenous wastes are important. Plants does not have specific organs to excrete these wastes. Plants break down waste substances at much slower rate than in animals. Hence accumulation of waste is also much slower. Green plants in darkness and plants that do not possess chlorophyll produce carbon dioxide and water as respiratory waste products. Oxygen itself can be considered as a waste product generated during photosynthesis, that exits out side through stomata of leaves and lenticels of stem.

- *How do the plants manage or send out waste products from its body?*

Plants can get rid of excess water by a process like transpiration and guttation. Waste products may be stored in leaves, bark, and fruits. When these dead leaves, bark, and ripe fruits fall off from the tree then waste products in them are get rid off. Waste gets stored in the fruits in the form of solid bodies called Raphides. However several compounds are synthesized by the plants for their own use especially for defence. Many plants synthesize chemicals and store them in roots, leaves, seeds, etc., for protection against herbivores. Most of the chemicals are unpleasant to taste. Hence, herbivores usually do not prefer to eat such plants. Some of the chemicals are toxic and may even kill the animal that eats them.

Think and discuss

- Why are weeds and wild plants not affected by insects and pests?

Some of the plants secrete chemicals when injured. These chemicals seal the wound and help the plant to recover from an injury. Some of the plants release attractants for other organisms which will help the plants for pollination, seed dispersal or even in their nutrition. For example, plants having root nodules secrete chemicals to attract Rhizobia into the surroundings of the roots and form a symbiotic relationship with the rhizobium. These compounds are called secondary metabolites.

- *Why plants shed their leaves and bark periodically?*

The biochemical substances produced in plants are of two types, primary metabolites and secondary metabolites. The materials like carbohydrates, fats and proteins are primary metabolites. The materials which do not require for normal growth and development are secondary metabolites. e.g.: Alkaloids, Tannins, Resins, Gums, and Latex etc. Though plants produce these chemical for their own use. Man found the usage of these chemicals for own benefits. They are generally coloured and fragrant.

Alkaloids:

These are nitrogenous byproducts and poisonous. These are stored in different parts of the plants. Common alkaloids in plants and their uses are given below.

Table-5

ALKALOID	PLANT	PART	USES
Quinine	*Cinchona officinalis* (Cinchona)	Bark	Antimalarial drug
Nicotine	*Nicotiana tobacum* (Tobacco)	Leaves	Insecticide, stimulant
Morphine, Cocaine	*Papaver somniferum* (Opium)	Fruit	Pain killer
Reserpine	*Rauwolfia serpentina* (Snake root)	Root	Medicine for Snake bite,High BP
Caffeine	*Coffea arabica* (Coffee plant)	Seed	Central nervous system Stimulant
Nimbin	*Azadirachta indica*(neem)	Seeds, Barks, Leaves.	Antiseptic
Scopolamine	*Datura stramonium*	Fruit, flower	Sedative
Pyrethroids	*Chrysanthemum sps*	Flower	Insecticides

Papaver *Rauwolfia* *Coffea arabica* *Tobacco* *Datura*

Fig-11: Plants which produce Alkaloids

- *Name the alkaloids which are harmful to us?*

Tannins: Tannins are carbon compounds. These are stored in different parts of the plant and are deep brown in colour. Tannins are used in tanning of leather and in medicines e.g. *Cassia*, *Acacia*.

Resin: Occur mostly in Gymnosperms in specialized passages called resin passages. These are used in varnishes- e.g.Pinus.

Fig-12(a): Cassia

Fig-12(b): Acacia

Fig-12(c): Pinus

Gums: Plants like Neem, Acacia oozes out a sticky substance called gum when branches are cut. The gum swells by absorbing water and helps in the healing of damaged parts of a plant. Gums are economically valuable and used as adhesives and binding agents, in the preparation of the medicines, food, etc.

Latex: Latex is a sticky, milky white substance secreted by plants. Latex is stored in latex cells or latex vessels. From the latex of *Hevea brasiliensis* (Rubber plant) rubber is prepared. Latex from *Jatropha* is the source of bio-diesel. Do you know which part of jatropha used in production of bio-diesel.

Fig-13(a): Neem

Fig-13(b): Jatropha

Fig-13(c): Rubber plant

Do you know?

Chewing gum is a type of sticky for chewing made dates back 5000 years. Modern chewing gum originally made of natural latex from plants like chicle, sapota etc. Whenever pollen grains enter in our body they cause allergy due to the presence of nitrogenous substances. These allergens cause skin allergy and asthma. Ex: *Parthenium*.

- *Do roots secrete?*

'Brugman' a botanist proved from his experiments that the roots not only absorb fluid from soil, but returns a portion of their peculiar secretions back into it. We can see such instances in plants like apple where a single apple crop for 4 or 5 years continuously in the same soil, that fail to produce fruits. It will not give proper yield even if you use lot of fertilizers.

- *Do you think there is any relation between reduction in yielding and root secretions?*
- *Why do we get peculiar smell when you shift the potted plants?*

Excretion Vs Secretion

Excretion and secretion are the same in nature. Since both are involved in passage or movement of materials. Both processes move and eliminate unwanted components from the body. Excretion is the removal of materials from a living being, while secretion is movement of material from one point to other point. So secretion is active while excretion is passive in nature. Humans excrete materials such as tears, urine, Carbon dioxide, and sweat while secretion on other hand, includes enzymes, hormones, and saliva. In plants too we find excretion through roots into its surroundings and falling off leaves and bark. Secretions occur in the plant body in form of latex, resins, gums etc.

Key words

Creatinine, tubular fluid, peritubular network, podocyte, glomerulus, PCT, DCT, afferent arteriole, efferent arteriole, calyces, micturition, urochrome, dialyser, heamodialysis, anticoagulant, alkaloids, biodiesel, loop of Henle.

What we have learnt

- Due to metabolism several harmful excretory products are formed and process of removing toxic waste from the body is called execretion.
- The human excretory system comprises a pair of kidneys, a pair of ureters, urinary bladder, urethra.
- Each kidney is composed of approximately 1.3 to 1.8 millions of uriniferous tubules or nephrons, which are structural and functional units of kidney.
- A nephron comprises glomerulus,bowman's capsule, proximal convoluted tubule (PCT), Henle's loop, Distal convoluted tubule (DCT) and collecting tubule.

- Formation of urine involves four stages. 1) Glomerular filteration, 2) Tubular Reabsorption, 3) Tubular Secretion, 4) Formation of concentrated Urine.
- Kidneys remove nitrogenous waste from body, maintains water balance (osmoregulation), salt content, pH, and blood pressure in human body.
- Dialysis machine is an artificial kidney which filters the blood to remove the metabolic wastes out side the body.
- Kidney transplantation is a permanent solution to renal failure patients.
- Different animals have different excretory organs e.g. amoeba-contractile vacuole, platyhelminthes- flame cells, annelida-nephridia, arthropoda-malpighian tubule, reptiles, birds and mammals-kidney.
- There are no special organs for excretion in plants. Plants store different waste materials in leaves, bark, roots, seeds which fall of from the plants.
- Plant metabolites are two types i) primary metabolites eg: proteins carbohydrates and fats. ii) secondary metabolites eg: alkaloids, gums, tannins, latex and resins. These are economically important to us.
- Excretion is the removal of material from living beings where as secretion is movement of materials from one point to other.

Improve your learning

Z2V9H4

1. What is meant by excretion? Explain the process of formation of urine. (AS1)
2. How are waste products excreted in amoeba?(AS1)
3. Name different excretory organs in human body and excretory material generated by them?(AS1)
4. Deepak said that 'Nephrons are functional and structural units of kidneys' how will you support him?(AS1)
5. How plants manage the waste materials?(AS1)
6. Why do some people need to use a dialysis machine? Explain the principle involved in it.(AS1)
7. What is meant by osmoregulation? How is it maintained in human body?(AS1)
8. Do you find any relationship between circulatory system and excretory system? What are they?(AS1)
9. Give reasons(AS1)

   A. Always vasopressin is not secreted.

   B. When urine is discharged, in beginning it is acidic in nature later it become alkaline.

C. Diameter of afferent arteriole is bigger than efferent arteriole.

D. Urine is slightly thicker in summer than in winter?

10. Write differences(AS1)

A. Functions of PCT and DCT
B. Kidney and artificial kidney
C. Excretion and secretion
D. Primary metabolites and secondary metabolites

11. There is a pair of bean-shaped organs 'P' in the human body towards the back, just above the waist. A waste product 'Q' formed by the decomposition of unused proteins in liver is brought into organ 'P' through blood by an artery 'R'. The numerous tiny filters 'S' present in organ 'P' clean the dirty blood goes into circulation through a vein 'T'. The waste substance 'Q' and other waste salts and excess water form a yellowish liquid 'U' which goes from organ 'P' into a bag like structure 'V' through two tubes 'W'. This liquid is then thrown out of the body through a tube 'X'.(AS1)

(a) What is (i) organ P and (ii) waste substance Q.
(b) Name (i) artery R and (ii) vein T
(c) What are tiny filters S known as?
(d) Name (i) liquid U (ii) structure V (iii) tubes W (iv) tube X.

12. The organ 'A' of a person has been damaged completely due to a poisonous waste material 'B' has started accumulation in his blood, making it dirty. In order to save this person's life, the blood from an artery in the person's arm is made to flow into long tubes made of substance 'E' which are kept in coiled form in a tank containing solution 'F'. This solution contains three materials 'G', 'H' and 'I' similar proportions to those in normal blood. As the person's blood passes through long tubes of substance 'E', most if the wastes present in it go into solution 'F'. The clean blood is then put back into a vein in the person for circulation. (AS1)

(a) What is organ A?
(b) Name the wastes substance B.
(c) What are (i) E, and (ii) F?
(d) What are G, H and I?
(e) What is the process described above known as?

13. Imagine what happens if waste materials are not sent out of the body from time to time?(AS2)

14. To keep your kidneys healthy for long period what questions will you ask a nephrologist/ urologist?(AS2)

15. What are the gum yielding trees in your surroundings? What procedure you should follow to collect gum from trees?(AS3)

16. Collect the information about uses of different kinds alkaloids, take help of Library or internet?(AS4)
17. Draw a neat labeled diagram of L.S of kidney?(AS5)
18. Describe the structure of nephron with the help of a daigram. (AS5)
19. Draw a block daigram showing the path way of excretory system in human beings. (AS5)
20. If you want to explain the process of filtration in kidney what diagram you need to draw.(AS5)
21. List out the things that makes you amazing in excretory system of human being.(AS6)
22. You read about 'Brain dead' in this chapter. What discussions would you like to have why you think so?(AS6)
23. We people have very less awareness about organ donation, to motivate people write slogans about organ donation?(AS7)
24. After learning this chapter what habits you would like to change or follow for proper functioning of kidneys?(AS7)

Fill in the blanks

1. Earthworm excretes it's waste material through ____________.
2. The dark coloured outer zone of kidney is called ____________.
3. The process of control of water balance and ion concentration with in organism is called________.
4. Reabsorption of useful product takes place in ____________ part of nephron.
5. Gums and resins are the __________ products of the plants.
6. Bowman's capsule and glomerulus taken together make a ___________.
7. The alkaloid used for malaria treatment is ________________.
8. The principle involved in dialysis is __________________.
9. Rubber is produced from ______________ of Heavea braziliensis of rubber plants.
10. ________________ performed first Kidney Transplantation.

Choose the correct answer

1. The structural and functional unit of human kidney is called ( )
   (A) Neuron (B)nephron (C)nephridia (D)flame cell
2. The excretory organ in cockroch ( )
   (A) malpighian tubules (B) raphids (C) ureters (D) nephridia